Accessible Teaching of Bayesian Statistics in JASP: Intuitive Examples of Biodiversity Estimation

Ziyuan Liu

Supervised by Prof. Dr. Eric-Jan Wagenmakers and MSc. Henrik Godmann

Psychological Methods Department

University of Amsterdam

Abstract

Frequentist statistics is dominant in education and research, but many students and even researchers misunderstand its key concepts (e.g., *p*-values and confidence intervals; Greenland et al., 2016). We suggest that this is partly because frequentist concepts are hard for beginners to comprehend. By contrast, the Bayesian framework may offer a more intuitive pathway into statistics. However, teaching Bayesian statistics to beginning students needs introductory-level materials. Therefore, we designed two JASP modules that introduce Bayesian statistics through an intuitive example of biodiversity estimation—estimating the number of different species on a fictional island. The static module introduces the fundamentals of Bayesian concepts in a straightforward way, while the full module extends the example by adding species abundance estimates. This project aims to bring Bayesian statistics to a younger audience, improving Bayesian literacy and promoting reform in statistical education.

Keywords: Bayesian Statistics, Statistical Education, JASP, Biodiversity Estimation

Accessible Teaching of Bayesian Statistics in JASP: Intuitive Examples of Biodiversity Estimation

Introduction

We teach it (frequentist statistics) because we use it; we use it because we teach it.

—Wasserstein & Lazar (2016)

Adapted from *the ASA Statement on P-Values*

Statistical Anxiety and Classical Solutions

Statistics plays a vital role in our daily lives. It forms the basis of nearly all quantitative sciences (Puklek Levpušček & Cukon, 2020) and is becoming increasingly significant in modern interdisciplinary fields like psychology (Lethbridge et al., 2024; Meng, 2009). However, many students find statistics challenging and experience statistics anxiety (Onwuegbuzie & Wilson, 2003; Silva & Sarnecka, 2025). Studies have revealed that being anxious about statistics brings a wide range of negative effects, such as low confidence in the subject (Hanna & Dempster, 2009; Hunt et al., 2023), procrastination (Dunn, 2014; Lethbridge et al., 2024; Trassi et al., 2022), and poor academic performance in the subject and overall (Lethbridge et al., 2024; MacArthur & Santo, 2023). Therefore, helping students engage with statistics more effectively is crucial both for the statistical education and for the personal development of the students.

Understanding why students find statistics difficult can inform better course designs and improve learning outcomes. Research highlighted that the obstacles for introductory-level statistics courses are: knowledge is abstract (Henriques, 2016; Silva & Sarnecka, 2025), lectures are too mathematical (Meng, 2009; Tishkovskaya & Lancaster, 2012), and students do not know much mathematics (Bromage et al., 2022; Tishkovskaya & Lancaster, 2012). These insights informed suggestions on course design, including making courses more practical (Bromage et al., 2022), using motivational techniques (Tishkovskaya & Lancaster, 2012), providing examples (Lovett & Greenhouse, 2000), and employing software (e.g., JASP) to visualise and simulate data (Bromage et al., 2022; Garfield, 1995). Although these are valuable suggestions, few reflections shed light on knowledge itself – it seems that learning statistics "must" be challenging.

Bayesian Statistics as a Better Framework

The present study, along with some other authors (Sedgwick et al., 2022), proposes that students struggle with statistics also because the widely used frequentist framework is counterintuitive and that most students, as well as some researchers, do not understand it (Greenland et al., 2016; Szucs & Ioannidis, 2017). To illustrate the point, consider the logic of frequentist statistics. We first hypothesise that the null hypothesis (H_0) is true, which means that there is no effect. Based on the assumption, we obtain a "sampling distribution" for all possible outcomes of the test statistic (e.g., t-value) had we repeated the experiment infinitely many times. Then, we estimate the probability of obtaining data at least as extreme as actual data under H_0, generating a value of p. Finally, we compare the p-value with some threshold (α) to decide whether we reject H_0 (i.e., draw a conclusion that there is "some" effect). Learning frequentist logic is challenging for beginners for two reasons. Firstly, beginners find it hard to relate to the sampling distribution since it involves the hypothetical scenario of repeating a process infinitely. Secondly, the frequentist framework first assumes that H_0 is true and then evaluates the plausibility of the observed data (i.e., $p(D \mid H_0)$, where D is for the data). This goes against our intuition, which tends to evaluate the probability that our hypotheses are true given the data, rather than the other way around.

By contrast, we believe that the Bayesian framework is more accessible to students. The principle of Bayesian inference is that our posterior knowledge (i.e., what we know now) is the product of our prior knowledge (i.e., what we knew before) and the evidence from the data (i.e., what we learnt), which aligns perfectly with our daily experience – we learn from the past for the future. Besides, Bayesian statistics intuitively addresses the goal of hypothesis testing: measuring how likely different hypotheses are given our observed data (Etz & Vandekerckhove, 2018; Wagenmakers et al., 2018). Unlike frequentist statistics, which first assumes that a hypothesis is true, Bayesian statistics evaluate our hypotheses by measuring how well they predict our data. The better a hypothesis predicts the data, the more credible it becomes. With straightforward logic, the conclusions from Bayesian statistics are also easy to understand. For example, the

Bayes factor (BF) directly means “the data are x times more likely to occur under one hypothesis than another”, which we believe is completely comprehensible even for high school students. Many scholars also appreciate the intuitive nature of Bayesian statistics and regard it as the “Theory of Common Sense” (Wagenmakers & Matzke, 2024).

Besides being counterintuitive, the logic of frequentist statistics also impedes reproducible research and logical reasoning. Taking into account the strength of the evidence, frequentist statistics allow weak evidence to be considered “significant”. Scientists have identified that a p-value just below 0.05 represents only “suggestive” evidence, and allowing such results to imply conclusions can lead to the replication crisis (Johnson, 2013; Nuzzo, 2014). Considering logical thinking, frequentist statistics promotes asymmetrical scientific reasoning, since we cannot quantify the evidence for H_0, but only the evidence against it (Dienes & Mclatchie, 2018; Wagenmakers et al., 2018). In other words, it cannot distinguish “evidence of absence" from "absence of evidence” (Altman & Bland, 1995). Furthermore, frequentist statistics has led people to care only about significance, not effect size (Dienes & Mclatchie, 2018; Nuzzo, 2014). This may cause people to trust findings that are statistically significant but practically meaningless.

The Bayesian framework effectively addresses these problems. Firstly, without significant thresholds, the Bayes factor quantifies the strength of the evidence directly (Stern, 2016; Wagenmakers et al., 2018), and the authors are encouraged to interpret it as its number (Wagenmakers & Matzke, 2024). Secondly, Bayesian statistics promotes symmetric reasoning, giving H_0 a fair chance to gain support (Dienes & Mclatchie, 2018; Etz & Vandekerckhove, 2018; Wagenmakers et al., 2018), so researchers can also draw a “no effect” conclusion. Thirdly, in Bayesian analysis, researchers use priors, which are specified a priori, to inform posteriors (Baig, 2020). Thus, researchers must be clear about which hypotheses they are testing using prior knowledge (Wagenmakers et al., 2018). With these advantages combined, we believe that the Bayesian framework offers a more robust and logical approach to statistics.

In summary, the frequentist framework presents two major challenges: it is counterintuitive for students to learn, and it has limitations that can undermine the rigour of

science. In comparison, the Bayesian framework is a more accessible pathway into the world of statistics because it is intuitive, robust, and scientific. Over the last few decades, more researchers have realised the advantages of Bayesian statistics and have started using it in practice (Bon et al., 2023; Hoegh, 2020; Rindskopf, 2020; van de Schoot et al., 2017). According to a systematic review, across six of the highest-profile psychology journals, the percentage of articles with Bayesian analyses has risen from almost 0% in 2004 to roughly 25% in 2024, and we expect it to approach 50% in 20 years (Pfadt et al., 2026).

Teaching and Using Bayesian Statistics: Status Quo and Future

However, despite the slow shift in focus, frequentist statistics remain dominant in science. So why are we still using the frequentist framework? According to the American Statistical Association (ASA), "we teach it (frequentist statistics) because it is what we do; we do it because it is what we teach" (Wasserstein & Lazar, 2016). While almost every single university teaches the frequentist framework in statistics courses, Bayesian education is very limited. According to Dogucu and Hu (2022), most top universities do not offer Bayesian courses at the undergraduate level, and those that do typically offer them only as elective courses for senior students, with more than two prerequisites. With such limited exposure to Bayesian methods, how can the next generation of researchers apply the framework and benefit from it?

Given this unsatisfactory status quo, the current study, along with some other researchers, suggests introducing Bayesian methods to younger students (e.g., high school or junior undergraduate levels) (Berry, 1997; Dogucu & Hu, 2022; Hoegh, 2020). As discussed above, young students can fully understand the Bayesian framework, benefit from its intuitive and logical nature, and avoid the transition from frequentist to Bayesian thinking, which can be challenging (Ferrari, 2022). However, we need suitable teaching materials for introductory-level Bayesian courses. Currently, most of the materials for Bayesian statistics fall into the following categories: (a) Books or book chapters introducing Bayesian statistics (e.g., Kruschke, 2014; McElreath, 2020) that are comprehensive but dense; (b) Articles theoretically or practically approaching the Bayesian framework (e.g., Baig, 2020; Depaoli et al., 2017; Schoot et al., 2021; Wagenmakers

et al., 2018) that are practical but advanced, and (c) Articles featuring sample Bayesian course designs (e.g., Berry, 1997; Dogucu & Hu, 2022) that are approachable but not detailed enough. These materials are not ready for use in introductory-level courses. According to previous recommendations, we propose that "ready-to-use" materials have the following features: (a) Engaging class examples and activities to capture student interest and demonstrate practical relevance (Lovett & Greenhouse, 2000); (b) Clear demonstrations and visualisations through easy-to-use statistical software (e.g., JASP) to make the technical parts of knowledge accessible (Bromage et al., 2022); (c) Manageable learning curve to satisfy students of different backgrounds, and (d) Appropriate instructions for students to master the thought process behind the knowledge.

Designing JASP Teaching Modules: Biodiversity Estimation

Guided by the previous discussion and the criteria we proposed, this project develops ready-to-use materials featuring JASP modules that introduce Bayesian statistics through interactive examples. The goal of our modules is to present the Bayesian framework in a way that is intuitive and accessible enough for absolute beginners. With specific learning goals, detailed explanations, and suggested interactive classroom activities, these modules are suitable both for teaching and for self-learning. Furthermore, with the help of JASP, which is a free and open statistical software (JASP Team, 2026), these modules are also fully open-access and easy to navigate.

Specifically, we focus on developing two modules under a single applicable example: biodiversity estimation. We invite students onto a journey towards an unknown, fictional island, with a task of estimating the number of different species and their abundance on it. The frequentist framework for biodiversity estimation typically relies on non-parametric estimators, including Chao1 (Chao, 1984), Chao2 (Chao & Chiu, 2016), Jack1 & 2 (Burnham & Overton, 1978), and some others, which are often too complex for beginners. However, this topic is also approachable within the Bayesian framework, since it essentially involves updating our beliefs through sampling. Furthermore, we believe that this topic makes great material for accessible

teaching of the Bayesian framework to beginners, given its popularity and importance in biological education, statistical education, and environmental protection (Bermudez & Lindemann-Matthies, 2020).

The general idea of the modules we built is to teach students Bayesian belief update through the realistic example of biodiversity estimation. By interacting with the JASP interface, users can specify priors for the hypothetical species count on the fictional island, sample animals on the island, and explore how the data change their beliefs.

However, to avoid overwhelming the students, we programmed two modules of increasing complexity. "Biodiversity Estimation (Static)" features a basic model that aims to teach the basic concepts of Bayesian statistics (e.g., prior, likelihood, and posterior) in the simplest settings. The model included some strong assumptions (see the next section for details) in order to minimise calculations and make it comprehensible for anyone with basic maths knowledge. "Biodiversity Estimation (Full)" features a follow-up model that relaxed some of the assumptions in the static model to make it more realistic. It is suitable for students who have some basics in Bayesian statistics, or have fully engaged with the static model. Both models are placed under the group of "Learn Bayes" and are easy to navigate with clear section divisions, detailed explanations, and informative tables and plots.

Together, these two modules and the relevant materials have significant implications for both students and lecturers. For introductory-level students, these modules help them understand the Bayesian framework and how to apply it to real-world problems, namely biodiversity estimation. For lecturers, our materials serve as a ready-to-use course design for Bayesian statistical courses. We consider our project a valid step towards bridging the gap between the Bayesian framework and junior-level statistical education, thereby promoting reform in statistical practices, empowering future scientists, and cultivating better research practices across scientific fields.

Module 1: Biodiversity Estimation (Static)

Overview and Objectives

The Biodiversity Estimation (Static) module aims to teach the principles of Bayesian belief update in a way that is completely accessible to beginners. When engaging with the model, users act as biologists who want to find out how many different species there are on the fictional island. Under the Bayesian framework, users should have some prior belief about the species count on the island, and the belief is shaped from prior to posterior by the data they observe, which in this case are the animals obtained from random sampling.

The module is split into multiple sections: prior, data, likelihood, posterior, and posterior prediction. Each section represents one key element in Bayesian statistics and is paired with informative explanations, tables, and plots. The central idea of this design is to unpack the process of Bayesian belief update by explaining one element at a time, so that users do not get confused.

In order to keep the calculation and logic as simple as possible, the mathematical model used in this module involves some strong assumptions. Firstly, the maximum number of species is set to 3, and the candidate species are set to "cat, horse, and pigeon" (i.e., there could be one of these three species, or there could be any combination of these three species). Secondly, the abundance of species is set to uniform, which means that if there are multiple species on the island, each species has the same abundance, which is not updated by the observed data. Thirdly, we assume that there is an infinite number of animals on the island, so that our observations are independent of each other and do not change the abundance of species. The impacts of these assumptions and how they help simplify the calculations will be discussed in the next subsection. With a thorough understanding that these assumptions could make the model less realistic, we think they are necessary to lower the cognitive load of the beginners in terms of mathematics, so that they can focus more on grasping the central idea of Bayesian statistics. We will relax some of these assumptions in the second module, where students will engage in a more complex scenario.

To help users better engage with the module, we propose the following learning goals. We think that, after fully engaging with the static module, students should be able to: (a) understand

the basic concepts of Bayesian statistics – prior, likelihood, and posterior; (b) understand how the Bayesian framework brings prior to posterior through likelihood; (c) use Bayesian statistics to compare hypotheses and interpret results; and (d) evaluate the posterior prediction and how it differs from the prior belief. Furthermore, we encourage students to think critically about the mathematical model in use. For example, is the model realistic enough? If not, what are the assumptions we made that are unrealistic in the real-world scenario?

Theoretical and Mathematical Framework

In this part, we will explain the static model in mathematical detail. Note that the point of this part is to fully unpack the logic "behind the scenes", and beginner students do not need to know every single detail explained in this part. Readers can refer to the next section for an example walk-through that includes some simple ideas for course design.

Bayes' rule

Bayes' rule is derived from the basic rules of probability (Gelman et al., 2003; Wagenmakers & Matzke, 2024). According to the rule of conditional probability (Kolmogorov, 1933), we have:

$$P(A \cap B) = P(A \mid B) \times P(B) \tag{1}$$

$$P(B \cap A) = P(B \mid A) \times P(A) \tag{2}$$

Since both $P(A \cap B)$ and $P(B \cap A)$ represent the probability of A and B occurring together, we have:

$$P(A \cap B) = P(B \cap A) \tag{3}$$

Therefore,

$$P(A \mid B) \times P(B) = P(B \mid A) \times P(A) \tag{4}$$

We then divide both sides of equation (4) by $P(B)$, giving us the following.

$$P(A \mid B) = P(A) \times \frac{P(B \mid A)}{P(B)} \tag{5}$$

which is the basic form of Bayes' rule.

In statistical testing, A often represents our belief in an unknown aspect of the world, and is written as θ, and B represents the data we observed, which is written as data, or D. Therefore, we can rewrite Bayes' rule as follows:

$$P(\theta \mid D) = P(\theta) \times \frac{P(D \mid \theta)}{P(D)} \tag{6}$$

On the left-hand side of the above equation, $P(\theta \mid D)$ translates to "the probability of θ being true given the observed data", which is called the "posterior belief". On the right-hand side of the equation, $P(\theta)$ is the probability of θ before seeing the data, or the "prior probability". $\frac{P(D|\theta)}{P(D)}$ is the "predictive updating factor", defined as the probability of observing the data (D) under the state of θ divided by the probability of observing the data in any possible state. To put it simply, the predictive update factor quantifies the relative predictive performance of θ and ensures that θ that better predicts the data has a larger posterior.

However, the probability of observing the data, $P(D)$, is often difficult to quantify directly. According to the law of total probability, it is defined by the following equation:

$$P(D) = \int P(D \mid \theta) P(\theta) \, \mathrm{d}\theta. \tag{7}$$

which translates into "the sum of the probability that the data occur in each possible state of θ".

Fortunately, in the scenario of hypothesis testing, we are more interested in "how much more credible one hypothesis is compared to another". Therefore, if we have two hypotheses, H_1 and H_2, we have the following:

$$P(H_1 \mid D) = P(H_1) \times \frac{P(D \mid H_1)}{P(D)} \tag{8}$$

$$P(H_2 \mid D) = P(H_2) \times \frac{P(D \mid H_2)}{P(D)} \tag{9}$$

Dividing (8) by (9) gives us the odds form of Bayes' rule:

$$\frac{P(H_1 \mid D)}{P(H_2 \mid D)} = \frac{P(H_1)}{P(H_2)} \times \frac{P(D \mid H_1)}{P(D \mid H_2)} \tag{10}$$

The equation does not require calculating $P(D)$, and is extremely powerful in statistics.

The following sections will apply Bayes' rule to our example of biodiversity estimation to further specify the model used in our "Biodiversity Estimation (Static)" module.

Hypotheses and Prior Belief

As we want to estimate how many different species there are on the island, our hypotheses are the different numbers of species on the island. As discussed above, to simplify the model, we restrict the maximum possible number of different species on the island to three, and we know that the candidate species are cat, horse, and pigeon. Therefore, we have two different levels of hypotheses. On the level of "species count", we have three hypotheses: there is only one (H_1), two (H_2), or all three (H_3) candidate species on the island. On the "species combination" level, each of these hypotheses is broken into sub-hypotheses. For example, under the hypothesis that there is only one species on the island (H_1), we have three sub-hypotheses: that the species is cat ($H_{1\alpha}$), horse ($H_{1\beta}$), or pigeon ($H_{1\gamma}$). The detailed specification of different hypotheses is shown in Table 1.

When specifying priors, the species count level priors are specified directly by the user. Then, the prior of each hypothesis is equally divided by the number of sub-hypotheses within the given hypothesis, since there is no additional information to prefer a specific combination of species. For example, since H_1 has three sub-hypotheses $H_{1\alpha}$, $H_{1\beta}$, and $H_{1\gamma}$, we have:

$$P(H_{1\alpha}) = P(H_{1\beta}) = P(H_{1\gamma}) = \frac{1}{3} \times P(H_1) \tag{11}$$

Likelihood

The likelihood describes "how likely the data are under each hypothesis". Here, the calculation differs between the species count level and the species combination level.

Calculating the likelihood at the species combination (i.e., sub-hypothesis) level is more straightforward than at the species count level. Since our model assumes independent sampling, with only one observation allowed at a time, the overall likelihood for a hypothesis is the product of the likelihoods of each observation under that hypothesis. Mathematically, let H be the hypothesis under investigation and let $\{A_1, A_2, \ldots, A_k\}$ be the different animals observed in the

sample. Then, the likelihood for H is defined as

$$P(D \mid H) = P(\{A_1, A_2, ..., A_k\} \mid H) = \prod_{i=1}^{k} P(A_i \mid H) \tag{12}$$

As the model assumes that the probability of capturing each individual animal is equal across individuals, the probability of observing each species is numerically the same as its abundance. Therefore, the key to calculating the likelihood is to define the abundance of different species in each hypothesis. As discussed above, to simplify the calculations, the model assumes equal abundance among the hypothetical species in each hypothesis, and the species abundance stays "static" (i.e., it does not update in response to the observed data). Thus, the abundance of each species under each hypothesis is shown in Table 1.

Table 1

The static model assumes equal and non-updating abundance for each species in each hypothesis.

Hypothesis	Sub-hypothesis	Species combination	Abundance		
			Cat	Horse	Pigeon
H_1	$H_{1\alpha}$	cat	1	0	0
	$H_{1\beta}$	horse	0	1	0
	$H_{1\gamma}$	pigeon	0	0	1
H_2	$H_{2\alpha}$	cat, horse	0.5	0.5	0
	$H_{2\beta}$	cat, pigeon	0.5	0	0.5
	$H_{2\gamma}$	horse, pigeon	0	0.5	0.5
H_3	H_3	cat, horse, pigeon	0.33	0.33	0.33

After obtaining the likelihood for each sub-hypothesis, we can calculate the likelihood for the hypotheses on the species count level (i.e., the probability of obtaining the observed data given the hypothesis that there are n different species on the island). According to Jeffreys (1961), we can use the following formula to calculate the marginal likelihood of a hypothesis when it has

multiple sub-hypotheses:

$$P(D \mid H) = \frac{P(D \cap H)}{P(H)} = \frac{\sum_i P(D \mid H_i)\, P(H_i)}{\sum_i P(H_i)} \tag{13}$$

Though it may seem complex at first, this equation simply translates to "the likelihood of a hypothesis equals the weighted average of the likelihood of each of its sub-hypotheses".

As we have obtained the likelihood of each hypothesis, we may also want to compare our hypotheses to see which one predicts the data better by how much. In Bayesian statistics, we do this by calculating the ratio of the likelihoods of two rival hypotheses, giving us a number that is called the "Bayes factor" (BF). Mathematically, if we compare two hypotheses H_1 and H_2, we have the following:

$$BF_{12} = \frac{P(D \mid H_1)}{P(D \mid H_2)} \tag{14}$$

Switching H_1 and H_2 in the above equation gives us BF_{21}, which is equal to $\frac{1}{BF_{12}}$. Interpreting the Bayes factor is straightforward, as it simply means that "the data are BF_{12} times more likely under H_1 than under H_2".

In summary, we first calculate the likelihood for the hypotheses at the species combination level and then combine them to calculate the likelihood at the species count level, which is what we are truly interested in. We can also compare hypotheses by taking the ratio of likelihoods of the hypotheses, giving us the Bayes factor. As introduced before, we made some assumptions when calculating the likelihood, which include (a) observations being sampled independently; (b) equal probability of capturing each individual animal; and (c) equal abundance across all candidate species in each hypothesis. We propose that these assumptions help keep the calculations simple, so that students can focus more on the general idea of Bayesian statistics.

Posterior Belief and Prediction

The posterior belief means "what we now believe about the state of the world given our prior and what data indicate". As stated in Bayes' rule (Equation (6)), the posterior of a hypothesis ($P(H \mid D)$) is the normalised product of its prior ($P(H)$) and likelihood ($P(D \mid H)$). The normalisation constant $P(D)$ is the probability of obtaining the observed data and is the sum

of the product of the prior and the likelihood for each hypothesis. Mathematically,

$$P(D) = \sum_i P(D \mid H_i) \times P(H_i) \tag{15}$$

When comparing two hypotheses, we can also use the odds form of Bayes' rule (Equation (10)) to avoid calculating $P(D)$. The odds form of Bayes' rule translates into "posterior odds = prior odds $\times$ likelihood ratio (i.e., the Bayes factor)". As the prior odds are specified by the user directly and the Bayes factor can be calculated, we can easily calculate the posterior odds for each pair of hypotheses by simple multiplication. To interpret the posterior odds, we can simply state: "Given our prior belief and the observed data, one hypothesis is x times more credible than the other."

After obtaining the posterior belief, we may also want to predict what we will observe in the next sample. According to the law of total probability, we can calculate the posterior predictive probability of each species s by taking the weighted average of its predictive probability across all possible hypotheses, weighted by their posterior probabilities. Mathematically,

$$P(s \mid D) = \sum_i P(s, H_i \mid D) = \sum_i P(s \mid H_i) \times P(H_i \mid D) \tag{16}$$

The term $P(s \mid H_i)$ is the probability of observing a species s under H_i, which is equal to the abundance of s given H_i. The term $P(H_i \mid D)$ is the posterior probability of hypothesis H_i. Having specified or calculated all these terms, we can easily calculate the predictive probability for each of the three candidate species.

Having established the mathematical framework for the static module, the following section demonstrates how these calculations are applied in practice through a step-by-step walk-through of the JASP module. Note that students, especially beginners, do not need to understand all the mathematical details explained in this section because what matters more to them is to engage with the module and try out different specifications, as the following walk-through will demonstrate.

Example Walk-Through

In this section, we put the model into practice by giving readers an example walk-through of the "Biodiversity Estimation (Static)" module in JASP so that they know what to expect. The section follows the structure of the JASP module, and we will also discuss some of the suggested ideas for classroom settings when demonstrating the module. Note that we have also added some introductory text in each part of the JASP module to help users navigate the module.

Prior Belief

The first section is the "Prior Belief" section. Here, users can specify a prior probability for each species count hypothesis (i.e., the island has one, two, or three different species). When specifying prior, users can choose between "equal probability" or "custom probability" for each species count. Selecting the former automatically assigns a uniform prior of 1/3 for each hypothetical species count; choosing the latter requires users to enter a customised number for each of the three possible species counts. The numbers are not restricted between 0 and 1, but the software automatically normalises the entered numbers into probabilities. In this example, we chose "custom probability", and entered "3", "6", and "1" respectively for species counts 1, 2, and 3. This gives us a prior belief as shown in Table 2. Our example prior indicates that we believe that two species is the most likely hypothesis, followed by one species, with three species being the least plausible. In classroom settings, teachers can explain the prior probabilities as "strength of initial belief before seeing the data", and can also encourage students to try out different priors, including extreme ones. It might also be helpful to ask students what their priors imply and how they would justify them.

After specifying priors on the species count level, the probability for each hypothesis is equally divided among its sub-hypotheses (i.e., the hypotheses on the species combination level). This gives us priors at the sub-hypothesis level, as shown in Table 3. It is a good exercise to help students understand this procedure by inventing some practice questions such as "how can you make the prior probability of each *species combination* hypothesis equal?". The answer is to make the priors on the species count level proportional to the number of sub-hypotheses it has,

Table 2

JASP can automatically normalise the priors entered by the user.

Hypothesis	Species count	Prior probability
H_1	1	0.300
H_2	2	0.600
H_3	3	0.100

which is 3:3:1 in the current scenario.

The JASP module allows table and plot displays for priors. Users can choose from species count or species combination as the level of hypotheses at which to display the prior. The prior table that resembles Table 2 or Table 3 is automatically available in the JASP results panel. By checking "Display prior plot", users can view the corresponding plot, shown in Figure 1.

Data

In the "Data" section, users can sample animals from the island. By clicking the "Sample an animal" button, a random individual animal is observed and recorded. Behind the scenes, JASP initialises a fictional island when the user opens the module, with some species and their abundances identical to one of the sub-hypotheses in Table 1. Clicking the button draws one sample with probabilities equal to each species' abundances. In this example walk-through, we sampled three times and observed a pigeon, a cat, and another pigeon (see Table 4 which reproduces the corresponding JASP output table). Thus, we have $D = \{pigeon, cat, pigeon\}$. Users can also click the "Reset" button to remove all previously observed animals and start over.

Likelihood

In the "Likelihood" section, users can explore the likelihood for each hypothesis on both the species count and the species combination levels.

When the user enables the "Table" checkbox, the software displays a table showing the likelihoods for both the current sample and the overall sample, and the numbers can be converted

Table 3

Prior probabilities for each species count hypothesis are equally divided to calculate the prior probabilities for each species combination hypothesis.

Hypothesis	Species count	Names of species	Prior probability
$H_{1\alpha}$	1	cat	0.100
$H_{1\beta}$	1	horse	0.100
$H_{1\gamma}$	1	pigeon	0.100
$H_{2\alpha}$	2	cat, horse	0.200
$H_{2\beta}$	2	cat, pigeon	0.200
$H_{2\gamma}$	2	horse, pigeon	0.200
H_3	3	cat, horse, pigeon	0.100

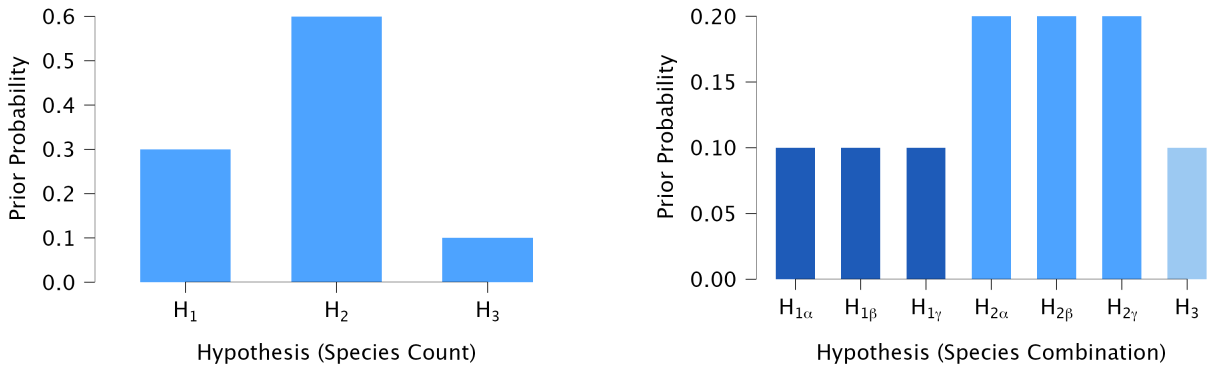


Figure 1

Users can choose to display prior plots on either the species count level (left panel) or the species combination level (right panel).

Table 4

The example walk-through features three samples: two previous samples (one pigeon and one cat) and one current sample (a pigeon).

Species	Previous samples	This sample	Total
cat	1	0	1
horse	0	0	0
pigeon	1	1	2

to logarithmic values since the likelihood shrinks quickly as we collect more sample. The overall likelihood is the product of the likelihood of each sample. For example, Table 5 describes the likelihood for each species combination level hypothesis, and Table 6 for each species count level hypothesis.

The results make sense mathematically. In Table 5, the sample likelihood for $H_{1\gamma}$ it is 1 because only pigeons are possible under this hypothesis, for $H_{2\beta}$ and $H_{2\gamma}$ it is 0.5 since the abundance of pigeon is 1/2 in these hypotheses, and for H_3 it is 0.333, which equals the abundance of pigeons under that hypothesis. The overall likelihood is the product of the sample likelihood for each sample. For all hypotheses under H_1, the overall likelihood is 0 since the data have two species. Under H_2, only $H_{2\beta}$ is possible overall because it can predict both cats and pigeons, with an overall likelihood of $(\frac{1}{2})^3$. H_3 is also possible with an overall likelihood of $(\frac{1}{3})^3$.

In Table 6, the overall likelihood is easy to understand, as it is the average likelihood of the sub-hypotheses for each hypothesis. The sample likelihood is more challenging to comprehend because it is the weighted likelihood of the sub-hypotheses by their posterior after the last sample, which is not shown directly in the table. Here, we do not expect students to understand all of the details, but they should be able to calculate the likelihood for the species combination level hypotheses and understand that the overall likelihood is the product of the likelihood for each sample.

Table 5

The species combination level likelihood after observing three samples (two pigeons and one cat).

Hypothesis	Species count	Species	Sample likelihood	Overall likelihood
$H_{1\alpha}$	1	cat	0.000	0.000
$H_{1\beta}$	1	horse	0.000	0.000
$H_{1\gamma}$	1	pigeon	1.000	0.000
$H_{2\alpha}$	2	cat, horse	0.000	0.000
$H_{2\beta}$	2	cat, pigeon	0.500	0.125
$H_{2\gamma}$	2	horse, pigeon	0.500	0.000
H_3	3	cat, horse, pigeon	0.333	0.037

Table 6

The species count level likelihood is the weighted average of the species combination level likelihood.

Hypothesis	Species Count	Sample Likelihood	Overall Likelihood
H_1	1	0.000	0.000
H_2	2	0.500	0.042
H_3	3	0.333	0.037

When the user ticks the "Plot" checkbox, the software visualises the likelihood through bar plots. Users can specify what data to display by selecting different levels of hypotheses and switching between "Last sample" and "Overall". The sample figures shown in Figure 2 demonstrate the overall likelihood for both levels of hypotheses. Furthermore, clicking "Hide impossible hypotheses" allows users to remove the hypotheses with zero likelihood from the plots to make them more tidy.

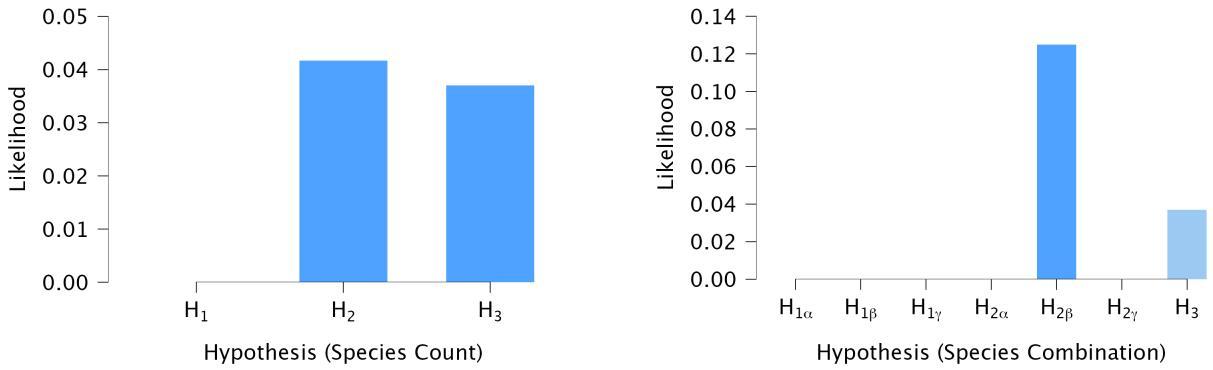


Figure 2

Users can show the likelihood for the last sample or overall, at either level of hypotheses.

Another important feature in the likelihood section is the Bayes factor. As discussed before, the Bayes factor is the ratio of the likelihoods of two hypotheses, which is commonly used when we compare how well these hypotheses align with the data. In order to enable this feature, users need to tick the "Bayes factor: compare likelihood" checkbox, and specify pairwise comparisons by moving hypotheses from the left box to the right box. The comparisons are available on both species count and species combination levels, but here we will use the former as an example. We specified two comparisons: H_1 vs. H_2 and H_2 vs. H_3. As shown in Figure 3, JASP returned a "pizza plot" with an explanation below for each comparison.



Figure 3

The "pizza plots" describe how surprising the data are under each hypothesis, and students can make sense of them by imagining dipping a finger into a pizza of two flavours.

Although the Bayes factor is already intuitive enough to interpret (as discussed before, we can put it as "the data are *BF* times more likely under one hypothesis than another"), we believe that the pizza plots are an effective way to further visualise the Bayes factor. As the name suggests, teachers can instruct students to imagine randomly dipping a finger onto a pizza with two flavours of different areas, and the surprise level of the observed data for a hypothesis is comparable to how surprised a person would feel if their finger landed on a particular topping. For example, in Figure 3, when comparing H_1 and H_2, the "pizza" only has one "flavour" because H_1 is impossible; when comparing H_2 and H_3, the part for H_2 is slightly larger than that for H_3, but most people would agree that the evidence is not decisive for either of these hypotheses. Note that the order of the hypotheses matters when calculating the Bayes factor. We prefer using pizza plots to visualise the likelihood because it is easier to grasp than a bare number.

The Bayes factor is a key part of Bayesian statistics, so it is important that students understand it. In classroom settings, teachers could encourage students to interpret the Bayes factor and pizza plots themselves and correct them when they make mistakes. It would also be

interesting to let students draw some pizza plots themselves and see how beginners intuitively perceive the strength of the evidence without introducing any established criteria.

Posterior Belief

In the "Posterior Belief" section, users can explore, in a Bayesian way, how the observed data update the prior belief. This section brings together the prior and the likelihood and is critical for helping students understand Bayes' rule.

As discussed in the previous section, for each hypothesis, the posterior belief is the product of its prior and likelihood, normalised across all hypotheses. For the prior, if we consider all data as a whole or if there is only one sample, the prior is specified initially by the user; if we only consider the last sample where there are multiple samples, the prior is the posterior after the previous sample. For the likelihood, it has been calculated in the previous section and is ready for use.

The first feature in this JASP section is the belief update table and plot. A sample belief update table is shown in Table 7. When producing the table, we ticked the "Table" checkbox in the "Belief Update" section, specified hypothesis type as "species count", and set evidence from as "overall". Users can also change these settings to show the species combination level posterior, or examine the prior and likelihood only for the current sample (i.e., the prior is the posterior before this sample). We designed the table in a way that is easy to understand: first we have the prior and the likelihood, then we multiply these numbers, and finally we normalise these products to get the posterior. The logic flows naturally and explains the mechanics of Bayes' rule. In classroom settings, we think this table is a good way to introduce Bayes' rule, since the calculation is unpacked one step at a time and the logic aligns with our daily experience: the observed data shift our belief from prior to posterior. It is also critically important to let students know or see for themselves that different prior beliefs can lead to dramatically different posterior beliefs.

Belief update plots are also available. Figure 4 shows a sample plot for the species count hypotheses with the grey bars as the initial prior and the blue bars as the posterior. We can see that the probability of H_1 has dropped to 0 because it could not predict the data, and both H_2 and

Table 7

Posterior is the normalised product of prior and likelihood.

Hypothesis	Species count	Prior	Likelihood	Prior × Likelihood	Posterior
H_1	1	0.300	0.000	0.000	0.000
H_2	2	0.600	0.042	0.025	0.871
H_3	3	0.100	0.037	0.004	0.129

H_3 gained probability. By putting the posterior bars next to the prior bars, users can easily compare prior and posterior.

Another feature in this section is the "Evidence Accumulation" plot. As shown in the sample plot (Figure 5), we put the number of samples on the *x*-axis, and the posterior probability for each hypothesis after observing a given number of samples on the *y*-axis of the plot. In this way, users can see what each sample brings to our belief. It is manifested that hypotheses with a higher probability get a higher starting point, but eventually, if we have an infinite amount of data, the data will overwhelm the prior. In classroom settings, it can be helpful for students to interpret the critical turning points in the evidence accumulation plot, helping them better understand the logic of Bayesian statistics. Also, discussing how much data we need to reach a satisfactory credible level for a hypothesis given different priors can help students understand that "extraordinary claims require extraordinary support from data".

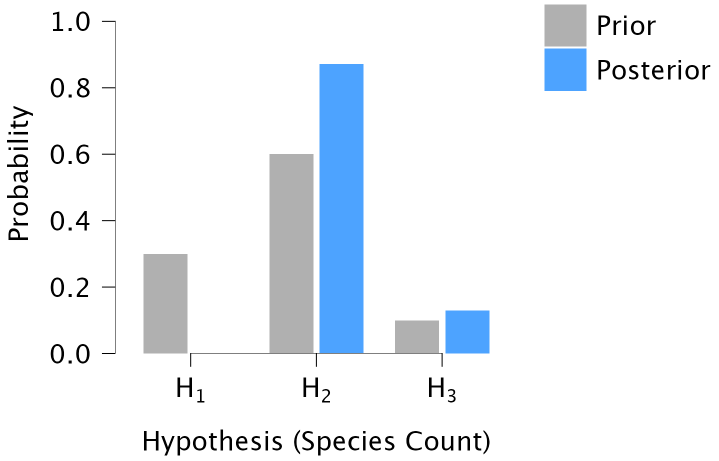


Figure 4

Hypotheses that predict the data poorly drop in probability, and those that predict the data well gain in probability.

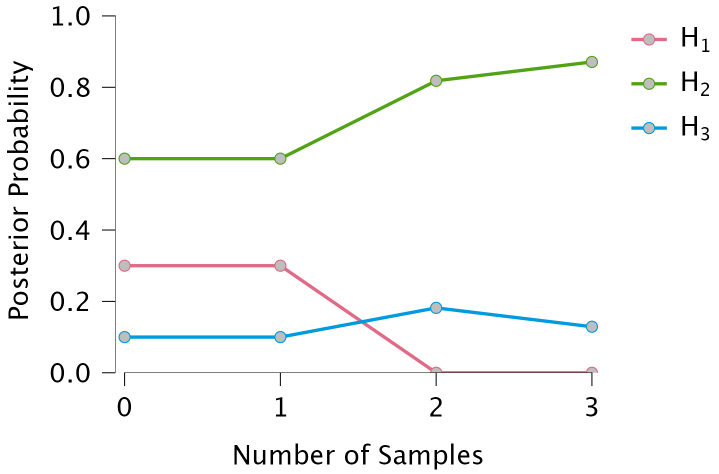


Figure 5

Evidence accumulates as we collect more samples.

The final feature of this section is "Compare Hypotheses". This is also an important feature since it works with the odds form of Bayes' rule. Recall that the odds form of Bayes' rule (Equation 10) demands that posterior odds be equal to prior odds times likelihood ratio (i.e., the Bayes factor), and since we already have all those probabilities, it is easy to calculate the posterior odds for any pair of hypotheses. Table 8 shows the posterior comparison table after specifying two comparisons in the GUI: H_1 vs. H_2 and H_2 vs. H_3. We specified the "Evidence from" field as "Overall", meaning that the table uses the initial prior and the overall likelihood to calculate the odds. We think the table is also completely understandable to beginners because it is all about simple multiplications (i.e., multiplying prior odds and likelihood ratio yields the posterior odds). It is good practice for students to interpret the columns in the odds table, for example, which hypothesis is more likely a priori, under which hypothesis the data are more likely, and which hypothesis is more credible given the prior and likelihood, by how much. These are all important questions to bring together what we have learnt so far. Also, students can try to calculate the posterior odds themselves given prior odds and likelihood ratio to practice using the odds form of Bayes' rule.

Table 8

The odds form of Bayes' rule is demonstrated by simply multiplying numbers in the posterior comparison table.

Hypothesis 1	**Hypothesis 2**	**Prior odds**	**Likelihood ratio** (BF)	**Posterior odds**
H_1	H_2	0.500	0.000	0.000
H_2	H_3	6.000	1.125	6.750

Posterior comparison plots (see Figure 6 as an example) are a great way to visualise the odds form of Bayes' rule. The figure contains three pizza plots with an explanation under each of them. These plots use the area of different parts to show the prior probability, likelihood, and posterior probability for different hypotheses, and are easier to make sense of than barely a number. To explain these plots to students, we can put it as "the area of each colour in the

posterior odds plot is the area of that colour in the prior odds plot times its area in the likelihood ratio plot".

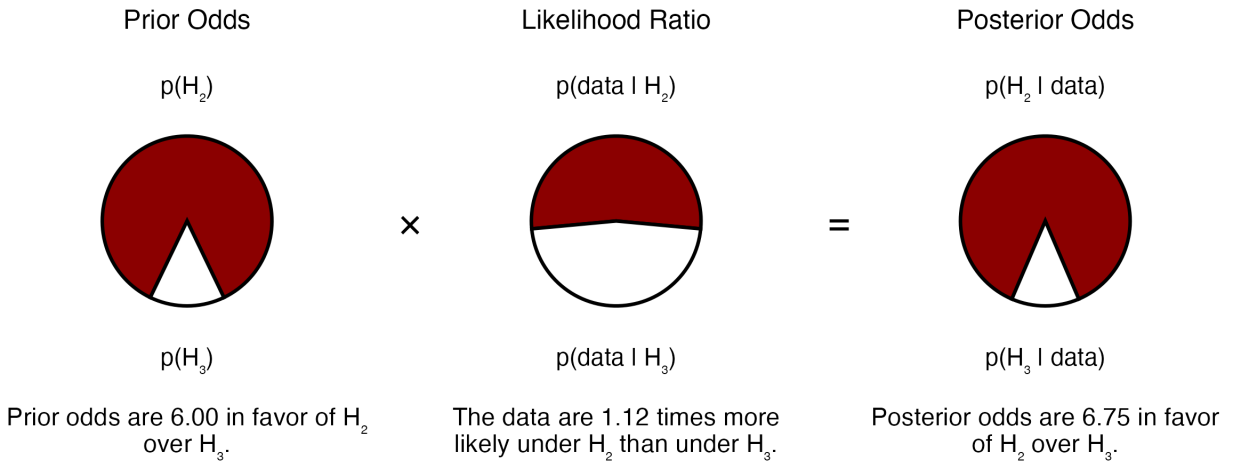


Figure 6

Combining the pizza plots for prior odds and likelihood gives us the pizza plot for posterior odds.

In summary, the "posterior belief" section is critical for introducing Bayes' rule to students and bring together all the "ingredients" of Bayesian statistics. Students who can understand the logic of this section have grasped the conceptual core of Bayesian statistics, and the remaining part, which is posterior prediction, is a natural follow-up step of the idea.

Posterior Prediction

After exploring the posterior belief, we also want to predict what we will observe for the next sample. This is the focus of the "Posterior Prediction" section. The main idea of this section is to showcase the posterior predictive probability of different species, or all seen species combined vs. other unseen species. Table 9 is a sample posterior predictive table. Note that it only shows hypotheses that have a posterior probability greater than 0. For each species, the posterior predictive probability is the sum across all hypotheses of the predictive probability of that species under each hypothesis, which is the product of the abundance of the species under that hypothesis and the posterior probability of the hypothesis (see Equation 16). For example, as

shown in Table 9, given our prior belief and after observing two pigeons and a cat, the probability of observing another cat is 0.478, and so is the probability of observing another pigeon; the probability of observing a horse is 0.043.

Table 9

Given the prior and the data, there is a fair chance of the next observation being a cat or a pigeon, but observing a horse is not that likely.

		Abundance			Joint probability		
Hypothesis	**Posterior**	**Cat**	**Horse**	**Pigeon**	**Cat**	**Horse**	**Pigeon**
H_2	0.871	0.500	0.000	0.500	0.435	0.000	0.435
H_3	0.129	0.333	0.333	0.333	0.043	0.043	0.043
Total					0.478	0.043	0.478

Posterior predictive plots are also available and are a great way to see how the observed data changed the predictive probabilities. For example, in Figure 7, the grey bars are the initial probabilities of certain observations, and the blue bars are the probabilities of those observations given our prior and the data. On the left panel, observations are classified by individual species, and on the right panel, they are classified by seen (i.e., cat and pigeon) vs. unseen (i.e., horse) species. We can see that before seeing any data, the probability of seeing each of the three species was uniform, but the data that contain two pigeons and a cat made cats and pigeons gain in probability, and horses lose in probability. Also, before seeing any data, all three species were unseen, making the predictive probability for unseen species equal to 1. However, after seeing the data, the probability of observing an old species has increased from 0 to about 0.956, which is close to 1.

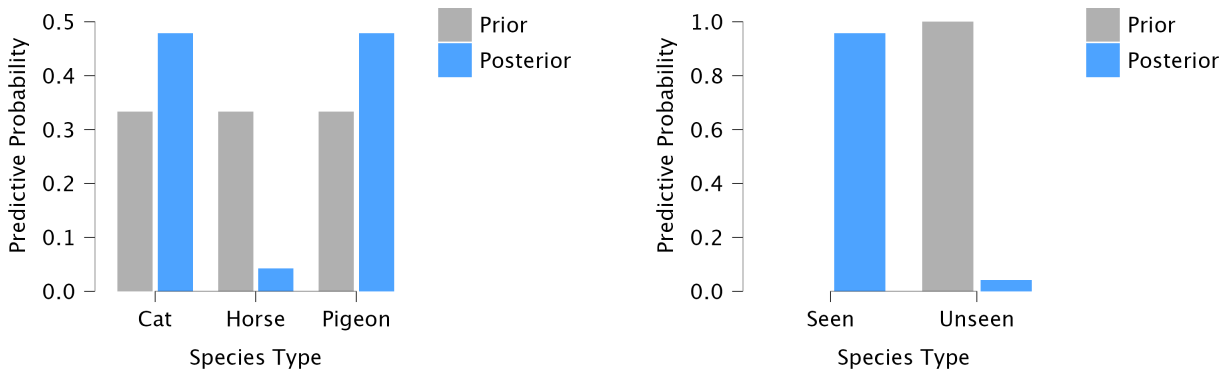


Figure 7

Our belief about what we will observe in the next sample has changed drastically after seeing the data.

The last feature in the "Posterior Prediction" section is the "Predictive change" plot (see Figure 8 as an example). These plots track the change of the posterior predictive probability for each species, or for seen and unseen species, as data accumulate. For example, Figure 8 shows that the posterior predictive probability for seen species increases rapidly as we collect more samples, and the species that appear in the sample increase in posterior predictive probability, while those that do not decrease. In classroom settings, these plots are useful for stimulating discussions about how many samples we need for a relatively stable and credible predictive probability. It is also important to let students understand that no matter how many samples we have collected, there are always chances to observe a new species that drastically changes our belief about the state of the world.

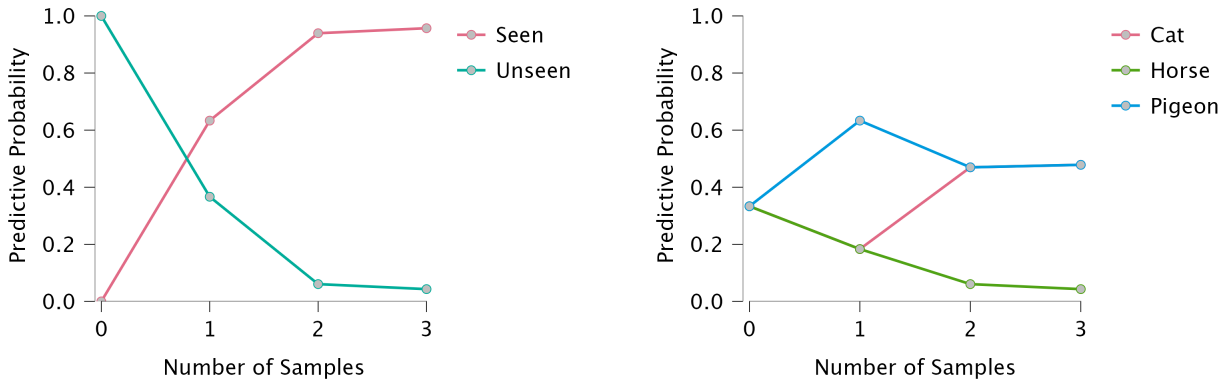


Figure 8

Each sample increases the probability of observing a species we have encountered before. However, there are always chances of observing something new in the next sample.

Here, we conclude our walk-through of the "Biodiversity Estimation (Static)" module, with the following short discussion reflected on the design of the static module.

Discussion

The aim of the static module is to teach the basic concepts of Bayesian statistics to beginners in an intuitive way. To achieve this goal, we designed the module with the following characteristics.

Firstly, we split the module into multiple sections, and each section aims to explain a single concept in Bayesian statistics. For example, the "Prior" section is all about prior belief, the "Data" section is about observing samples, and the "Likelihood" section is about computing likelihood. Not until the "Posterior Belief" section does the module involve the full form of Bayes' rule and bring the prior and likelihood together. This design makes it clear what each part of our module is trying to explain, and users can build up their knowledge one step at a time.

Secondly, we designed the module in a way that minimises calculations. From the start to the end, all that students need to understand is how to calculate likelihood and how to combine prior and likelihood to calculate posterior. We also made several assumptions to further simplify the calculation of the likelihood. We believe that minimising mathematical calculations can help

students focus more on the logic of Bayesian statistics, which is supposed to be simple and intuitive. Ultimately, Bayesian statistics is all about "posterior belief equals prior times likelihood".

Last but not least, we use the module to present the knowledge in a way that involves clear and informative tables and plots. The tables in our module are designed to be as self-explanatory as possible. For example, in the "Posterior Belief" section, the table presents the prior and the likelihood side by side, then their product, and finally the normalised posterior. This logic flow makes it easy for students to make sense of Bayes' rule and also for lecturers to explain it. The plots are all neat and tidy, with the fewest visual elements possible to convey the essential messages. For example, the prior and posterior plot has the posterior probability of each hypothesis plotted next to the prior probability, making it easier to compare. Also, the pizza plots for Bayes factor and belief update visualise comparisons between hypotheses vividly. In this report, we have also offered some ideas that may help lecturers explain these concepts to students.

However, as discussed before, the model used in the static module has various assumptions to simplify the scenario. Though those assumptions may ease students' burden and help them focus more on the basic concepts of Bayesian statistics itself, they also introduce limitations, making the module not realistic enough. Firstly, we assumed that we already knew the candidate species on the island (i.e., we restricted the maximum number of species on the island to three, and the candidate species are cats, horses, and pigeons). This is unrealistic because we never know exactly what to expect before going to an unknown island. Secondly, the abundance of each species is set to be equal when the hypothesis involves multiple species. This is also unrealistic because the abundance of species on an island is almost always uneven. For example, there could be thousands of ants but only two pandas on an island. The abundance is also static, which means that it is not learnt from the data. This results in equal posterior predictive probability for each species seen in the samples (see Figure 7 as an example), which does not make sense because, for example, we cannot justify equal predictive probability for cats and horses having observed 100 cats but only 5 horses in the samples. Last but not least, the module assumes that there is an

infinite number of animals on the island, while realistically, the total number of individuals on an island is always finite and is a function of the area of the island and species density.

Despite those limitations in the model, we still think that the static module serves as effective teaching material for beginners to learn the basics of Bayesian statistics. However, when using the module, users should be aware of those limitations, and it is also beneficial for lecturers to have students reflect on the model and discuss these limitations with them after they have fully engaged with the module. We believe that this discussion will provide a natural transition to the full module (covered in the next section), since the full module relaxes some of those assumptions and is more realistic than the static module.

Module 2: Biodiversity Estimation (Full)

Overview and Objectives

The Biodiversity Estimation (Full) module builds on the static module by relaxing some of its assumptions, making it more realistic. The background story is the same as the static module, where users try to find out how many different species there are on the fictional island. However, in this module, instead of giving each species a fixed abundance, users have an additional task, which is to estimate the abundance of each species on the island (i.e., Which species has the most individuals? What is the estimated proportion of each species?).

Like the static module, the full module is also divided into multiple sections, each section representing an element in Bayesian statistics. We also implemented informative JASP-style tables and plots to help students understand the content.

The full module differs from the static module in that it relaxes some of the latter's assumptions, making the mathematical model more complex but also more realistic. Firstly, the maximum number of species is now 10 (compared to 3 in the static module), and candidate species are no longer pre-specified, meaning any species may be encountered. Secondly, the species abundance is no longer set fixed, but it is learnt from the sample, with a uniform Dirichlet prior (see the next section for detailed explanations). However, we keep the assumption that the total number of individuals is infinite, to avoid overcomplicating the scenario. As with the

previous section, we will discuss how these model specifications come into play mathematically in the next section.

The full module is suitable for students who already have some basic understanding of Bayesian statistics and is best suited for those who have already engaged fully with the static module, since the two modules share the same underlying context. In addition to the learning goals for the static module, we propose two additional learning goals specifically for the full module. By engaging with the full module, we expect that students should be able to: (a) understand how to estimate the abundance of each species on the fictional island (in a Bayesian way) and (b) deepen the understanding of basic Bayesian concepts and be able to comprehend and calculate likelihood and posterior in a slightly more complicated scenario. Moreover, we also encourage students to reflect on the differences between the static module and the full module. For example, what impact does relaxing these assumptions have on making the model more realistic? Is the full module realistic enough for real-world biodiversity estimation? If not, what are its limitations?

Theoretical and Mathematical Framework

In this part, we will introduce the calculations behind the scenes of the full module. As we have already introduced Bayes' rule in the previous sections, we will dive straight into the model, starting with the prior belief and then introducing the calculations of species abundance, likelihood, posterior belief, and posterior prediction. Note that, just as with the static module, we do not expect students to understand all the mathematical details explained in this section. It is more important that students focus on how to estimate the abundance of species and how to make sense of and interpret the JASP output.

Hypotheses and Prior Belief

The prior in the full module means the same as in the static module. It refers to the belief about how many different species there are on the fictional island. However, there are some minor differences. Firstly, the full module does not have the species combination level prior. Unlike the static module, we do not know the candidate species in the full module, so it is impossible to

distribute the prior mass of each hypothetical species count across species combinations. Secondly, the maximum number of species is now 10, compared to 3 in the static module, making the situation more complex, yet more realistic. Thirdly, when specifying the prior, instead of entering the numbers for each species count, users will use distributions with different parameters. This design helps students become familiar with common distributions, which may also benefit their future studies.

Currently, four different types of distribution are available: the point distribution, the uniform distribution, the Poisson distribution, and the negative binomial distribution.

The point distribution represents the case where the user believes that only one species count (s_0) is possible and is given by:

$$P(S = s) = \begin{cases} 1, & s = s_0 \\ 0, & s \neq s_0 \end{cases} \tag{17}$$

This is rarely used because no one knows for sure how many species there are before stepping onto the island. Lecturers can demonstrate to students why putting all the prior mass into one single hypothesis is not a good idea.

The uniform distribution assigns equal probability to all possible species counts between the minimum (s_{min}) and the maximum (s_{max}), inclusive, entered by the user, and is given by:

$$P(S = s) = \begin{cases} \frac{1}{s_{max} - s_{min} + 1}, & s_{min} \leq s \leq s_{max} \\ 0, & s > s_{max} \text{ or } s < s_{min} \end{cases} \tag{18}$$

This is the go-to distribution when the user only has clues about the minimum and maximum possible number of species on the island without knowing anything else.

The Poisson distribution is another commonly used distribution for non-negative integers. The distribution is given by:

$$P(S = s) = \frac{\lambda^s e^{-\lambda}}{s!}, \quad s = 0, 1, 2, \ldots \tag{19}$$

The Poisson distribution has the following properties: Firstly, the distribution tends to be right-tailed, with the highest probability assigned to numbers around λ. Secondly, the distribution

has a single parameter, λ, which is both the mean and the variance of the distribution. In our module, the Poisson distribution matches the case when the user has some idea of the approximate number of different species on the island. The user can enter the approximate number as λ. The Poisson distribution can also be truncated in our module, with $s_{min} \geq 1$ and $s_{max} \leq 10$, and renormalised accordingly.

The negative binomial distribution is an "extended" version of the Poisson distribution. Unlike the Poisson distribution, which requires equal mean and variance (λ), the negative binomial distribution allows mean-dispersion parameterisation that can change the variance while maintaining the same mean. The distribution has two parameters μ and ϕ, and is given by:

$$P(S = s) = \frac{\Gamma(s+\phi)}{\Gamma(\phi)\, s!} \left(\frac{\phi}{\phi+\mu}\right)^{\phi} \left(\frac{\mu}{\phi+\mu}\right)^{s}, \quad s = 0, 1, 2, \ldots \tag{20}$$

with the following mathematical properties:

$$E[S] = \mu \tag{21}$$

$$\text{Var}(S) = \mu + \frac{\mu^2}{\phi} \tag{22}$$

As $\phi \to \infty$, $\frac{\mu^2}{\phi} \to 0$, and the distribution converges to the Poisson distribution. Users may find the negative binomial distribution useful when they have some idea of both the approximate mean of the number of species on the fictional island and their uncertainty about it. Truncation is also supported for the negative binomial distribution, with the lower bound being an integer no less than 1 and the upper bound no more than 10.

Together, these four types of distribution allow users to specify the prior in a convenient way, while also getting students familiar with some commonly used distributions. All distributions can be plotted in our module as users specify different parameters, and we will demonstrate an example of setting a prior in the next section.

Species Abundance

As mentioned before, a key objective of the full module is to teach students how to estimate species abundance from our data in a Bayesian way. Here, we work with the Dirichlet

distribution. For each hypothesis with a specific species count k larger than 1, let p_1, p_2, …, p_k denote the abundances of the k hypothetical species. Then, the probability density function is defined as

$$f(p_1,\ldots,p_k;\alpha_1,\ldots,\alpha_k) = \frac{1}{B(\alpha)}\prod_{i=1}^{k} p_i^{\alpha_i-1} \tag{23}$$

where $B(\alpha)$ is the normalising constant

$$B(\alpha) = \frac{\prod_i \Gamma(\alpha_i)}{\Gamma(\sum_i \alpha_i)} \tag{24}$$

and α_1, …, α_k are concentration parameters.

When focusing on a single species, the Dirichlet distribution reduces to the Beta distribution. For a given species i with an abundance of p_i, the probability density function of p_i is

$$f\left(p_i;\ \alpha_i,\ \sum_{j\neq i}\alpha_j\right) = \frac{1}{B\left(\alpha_i,\ \sum_{j\neq i}\alpha_j\right)}\, p_i^{\alpha_i-1}\,(1-p_i)^{\sum_{j\neq i}\alpha_j-1} \tag{25}$$

which is also expressed as $p_i \sim \text{Beta}(\alpha_i,\ \sum_{j\neq i}\alpha_j)$.

The only problem left is how to specify α_1, …, α_k, which is where a very useful property of the Beta distribution, conjugacy, comes into play. We first recall that in the Bayesian world, a central rule is $posterior \propto prior \times likelihood$. Under the hypothesis that there are k species on the fictional island, suppose that we have observed n_1, n_2, …, n_k individuals for species 1, 2, ..., k, then the prior for the abundance of species i (p_i) is given as:

$$f(p_i;\ \alpha_{i0},\ \alpha_{i0},\ldots,\alpha_{i0}) = \frac{1}{B\left(\alpha_{i0},\ \sum_{j\neq i}\alpha_{j0}\right)}\, p_i^{\alpha_{i0}-1}\,(1-p_i)^{\sum_{j\neq i}\alpha_{j0}-1} \tag{26}$$

For the likelihood, assuming independent sampling, we have:

$$P(n_1,n_2,\ldots,n_k \mid p_i) = \binom{\sum_j n_j}{n_i} p_i^{n_i}(1-p_i)^{\sum_{j\neq i} n_j} \tag{27}$$

By omitting all constants in the prior and likelihood and multiplying these terms together, we get the posterior:

$$f(p_i \mid n_1,n_2,\ldots,n_k) \propto p_i^{\alpha_{i0}-1}(1-p_i)^{\sum_{j\neq i}\alpha_{j0}-1} \times p_i^{n_i}(1-p_i)^{\sum_{j\neq i} n_j} \tag{28}$$

$$= p_i^{\alpha_{i0}+n_i-1}(1-p_i)^{\sum_{j\neq i}\alpha_{j0}+\sum_{j\neq i} n_j-1} \tag{29}$$

Here we notice that the posterior and the prior share the same kernel, and the posterior is also a Beta distribution:

$$p_i \mid n_1, n_2, \ldots, n_k \sim \text{Beta}(\alpha_{i0} + n_i, \sum_{j \neq i} \alpha_{j0} + \sum_{j \neq i} n_j) \tag{30}$$

The above expression tells us that we can easily update p_i by adding the number of observed individuals of species i (n_i) to the first parameter of its prior Beta distribution, and adding the combined number of observed individuals of all other species ($\sum_{j \neq i} n_j$) to the second parameter. Furthermore, if we start from an uninformative, uniform prior ($\alpha_1 = \alpha_2 = \ldots = \alpha_k = 1$), the posterior distribution of p_i is a distribution in which the first parameter is the total number of observed individuals of species i plus one, and the second parameter is the total number of observed individuals of all species other than i plus one. Conjugacy makes the updating process straightforward in classroom settings because students can estimate the posterior distribution of each species' abundance without doing complicated calculations, which is a reason why we use the Dirichlet distribution in this framework.

Likelihood

After obtaining the abundance of each species under each hypothesis, we can calculate the likelihood for each hypothetical species count (i.e., the biodiversity likelihood). Here, there are two things worth considering. Firstly, in the full module, observations are not restricted to one animal at a time. Instead, we allow multiple individuals in a single batch of data. Secondly, as in a real-world capture-recapture scenario, every time we capture an animal, we document it by putting a tag on it that does not affect the chance of it being recaptured, so that we can distinguish new individuals and recaptured individuals. Only new individuals are taken into account when calculating the likelihood.

We calculate the likelihood using a sequential updating strategy that accounts for one individual at a time. As sampling is independent, the total likelihood for the current batch of data is:

$$P(D \mid H) = \prod_{k=1}^{N} P(d_k \mid H) \tag{31}$$

where N is the total number of individuals in the sample and d_k denotes the k-th individual.

When calculating $P(d_k \mid H)$, the situation depends on whether we have seen the species of d_k before. If we have seen the species, then the likelihood is simply the abundance of the species of d_k under the given hypothesis in the current state of the world. For example, if d_k is a member of the species i with abundance (p_i) following the probability density function $p_i \sim \text{Beta}(\alpha, \beta)$, then we have

$$P(d_k \mid H) = \int_0^1 p_i f(p_i)\, dp_i = E[p_i] = \frac{\alpha}{\alpha + \beta} \tag{32}$$

which is easy to calculate without needing to perform any integration. If we have not seen the species, then we need to add another term to account for the fact that this individual could belong to any of the remaining unobserved species. This term equals the number of species not yet observed under the given hypothesis. For example, if the hypothetical species count is s, and we have observed s_{obs} different species prior to this sample, then the likelihood is

$$P(d_k \mid H) = \binom{s - s_{\text{obs}}}{1} \times \int_0^1 p_i f(p_i)\, dp_i = \binom{s - s_{\text{obs}}}{1} \times E[p_i] = (s - s_{\text{obs}}) \times \frac{\alpha}{\alpha + \beta} \tag{33}$$

In this way, we can calculate the likelihood for each individual in our sample sequentially and multiply them together for the overall likelihood of the sample. This likelihood is a key component when calculating the posterior belief in the following section.

Posterior Belief and Prediction

The posterior belief in the full module has two parts. The "biodiversity posterior" refers to the posterior belief about the number of different species on the fictional island. The "species abundance posterior" refers to our belief about the abundance of each species, given the prior and the observed data.

Calculating the biodiversity posterior is relatively straightforward. After specifying the prior ($P(H_i)$) and calculating the likelihood ($P(D \mid H_i)$) for each hypothetical species count, the posterior ($P(H_i \mid D)$) is the normalised product of the prior and likelihood across all possible hypothetical species counts. This is a direct application of Bayes' rule, and we have explained it in the previous section (see Equation 6 and Equation 15).

Calculating the species abundance posterior is a process of aggregating the species abundance distributions across hypotheses and making a combined distribution for each species. Here, we use Bayesian model averaging (Hoeting et al., 1999) to take the weighted average of them according to the posterior probability of each hypothetical species count. For example, let $f(p_j \mid H_i, D)$ denote the posterior density of the abundance of species j under hypothesis H_i, then:

$$f(p_j \mid D) = \sum_i f(p_j \mid H_i, D) \times P(H_i \mid D) \tag{34}$$

In addition to calculating the posterior belief, we may also want to predict the result of our next observation. Following the same structure as in the static module, we calculate the marginal probability of each species under all possible hypothetical species counts. Again, if $f(p_j \mid H_i, D)$ denotes the posterior density of the abundance of species j under hypothesis H_i, then the marginal probability of observing species j in the next sample is:

$$P(j \mid D) = \sum_i P(j, H_i \mid D) \tag{35}$$

$$= \sum_i \left[\int_0^1 p_j f(p_j \mid H_i, D)\, dp_j \right] \times P(H_i \mid D) \tag{36}$$

$$= \sum_i E[p_j \mid H_i, D] \times P(H_i \mid D) \tag{37}$$

Comparing Equation 34 and Equation 37 tells us that the predictive probability for species j is simply the mean of the marginal abundance distribution for that species, which is consistent with both statistical intuition and the underlying mathematics.

To sum up, the above section detailed what happened mathematically behind the scenes of our full module. Again, we do not expect students to understand every mathematical equation in perfect detail. It is more important for students to grasp the main idea of how to update our beliefs about both the number of different species on the fictional island and their abundance, in a Bayesian way, in a scenario that is slightly more complicated than the static module. As with the static module, readers can refer to the following section for an example walk-through of the full module with some suggested practical ideas for classroom settings.

Example Walk-Through

Just as with the static module, in this section, we will give users a step-by-step demonstration of our "Biodiversity Estimation (Full)" module in JASP. We will walk through our module one section at a time, and also discuss some suggested classroom activities as we progress.

Prior

In order to update our belief, we first need a "prior belief". In the full module, the prior belief is specified by using distributions with different parameters. In this example walk-through, we use a Poisson distribution with $\lambda = 6$, truncated to a minimum of 1 and a maximum of 10. The distribution has a mean of roughly 6, and the plot is shown in Figure 9. In addition, users can tick a checkbox to display an explanation of the current prior distribution.

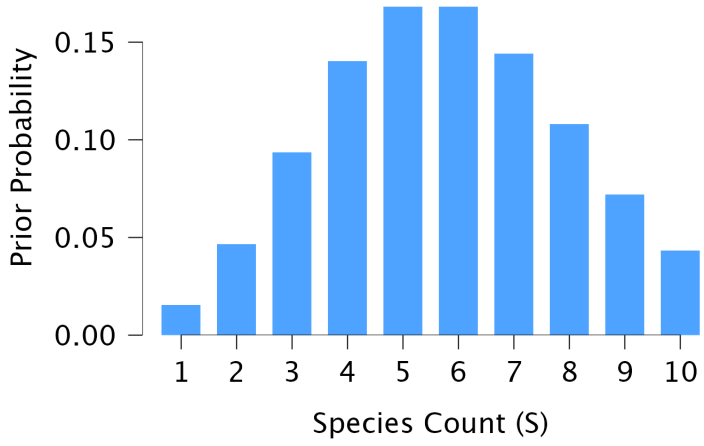


Figure 9

The Poisson distribution with $\lambda = 6$ reflects a prior belief that there are on average roughly 6 different species on the island.

A summary statistics table is also available (see Table 10 as an example). Here, users can choose what to display in the table by ticking different statistics in the user interface (i.e., the mean, the median, the mode, the standard deviation and the total probability that falls in a customised range).

Table 10

Users can customise the statistics shown in the prior summary statistics table.

Mean	Median	Mode	SD	$P(4 \leq S \leq 8)$
5.756	6.000	5.000	2.137	0.729

We believe that using distributions to specify prior belief not only simplifies the process but also gives students a chance to engage with these common distributions and become familiar with them. In classroom settings, we recommend lecturers not to focus too much on the mathematical form of those distributions but to visualise them with plots and let students tweak the parameters and truncations to see how they change the shape of the distributions. It is also a good exercise for students to verbalise their belief themselves from the distribution plot, or to transform their belief from words to an appropriate distribution.

Data

The data section is where the sampling comes into play. Here, we allow users to specify a sample size, which is the number of animals to observe in the next sample. In this example, our data contain three samples. The sample sizes are 5, 20, and 100, respectively. Note that when we observe those animals, we put a tag on them before letting them go, so that we can tell the individuals that we have already captured in previous samples from new individuals. Sample tables and graphs are available for both the current sample and all combined samples (see Figure 10 and Table 11 as examples). Unlike the static module, which is restricted to three candidate species (cat, horse, and pigeon), the full module allows for a broader, more general set of possible species. To ensure reproducibility, there is also an option called "Select an island", where the user can enter a number that serves as a seed to generate the fictional island. When this option is enabled, the species on the island stays the same even if the user clicks "Reset" to clear all the samples.

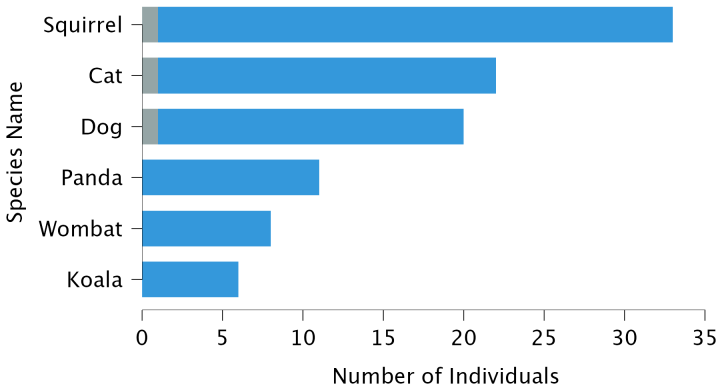


Figure 10

The bar plot for the current sample has different colours for the new and recaptured individuals.

Table 11

Recaptured animals are only counted once in the table and plot showing all samples combined.

Species	Number of individuals
Squirrel	37
Dog	27
Cat	25
Panda	13
Wombat	11
Koala	9
Total	122

Abundance

In the abundance section, we display estimates of the abundance of different species for each possible hypothetical species count. For example, across our three samples, we have already seen six different species, which are squirrel, dog, cat, panda, wombat, and koala. Thus, our estimates involve the abundance for each of these species under each of the hypotheses where $6 \leq S \leq 10$. As discussed above, we use the Dirichlet distribution to estimate the abundance of these species from our samples, which can be transformed into Beta distributions for each species under each hypothesis. The detailed estimation process has been introduced in the previous section and is not mandatory for introductory students. Figure 11 shows an example JASP plot for abundance estimation under the hypothesis that there are 6 species on the island. By interacting with the user interface, users can choose to display the mean, median, or mode for the prior and posterior distributions. Users can also see the parameter change of each Beta distribution by enabling the "Explain plots" checkbox. In the last sample of our example, we observed 97 new individuals, and 32 of them are squirrels, meaning that 65 are not. This moves the distribution of the abundance of squirrels from $\text{Beta}(6, 25)$ to $\text{Beta}(38, 90)$.

Abundance tables are also available with different estimates. For example, Table 12 shows the means of each species' abundance under different hypothetical species counts on the island. Both the table and the plot are organised in a way that species with more occurrences in the samples appear first.

In classroom settings, though we do not expect students to understand the mathematical details of how the distributions are updated, we do want to encourage them to appreciate the beauty of the Beta distribution. Firstly, the conjugacy of the Beta distribution allows us to update the parameters of the distributions simply by looking at the frequencies of each species in our samples. Secondly, we can calculate the mean of a Beta distribution by dividing the first parameter by the sum of both parameters. For lecturers, a good but challenging exercise would be to let students practise updating the species abundance by specifying a prior and looking at the frequency table for the species (for example, Table 11). In addition, students can benefit from

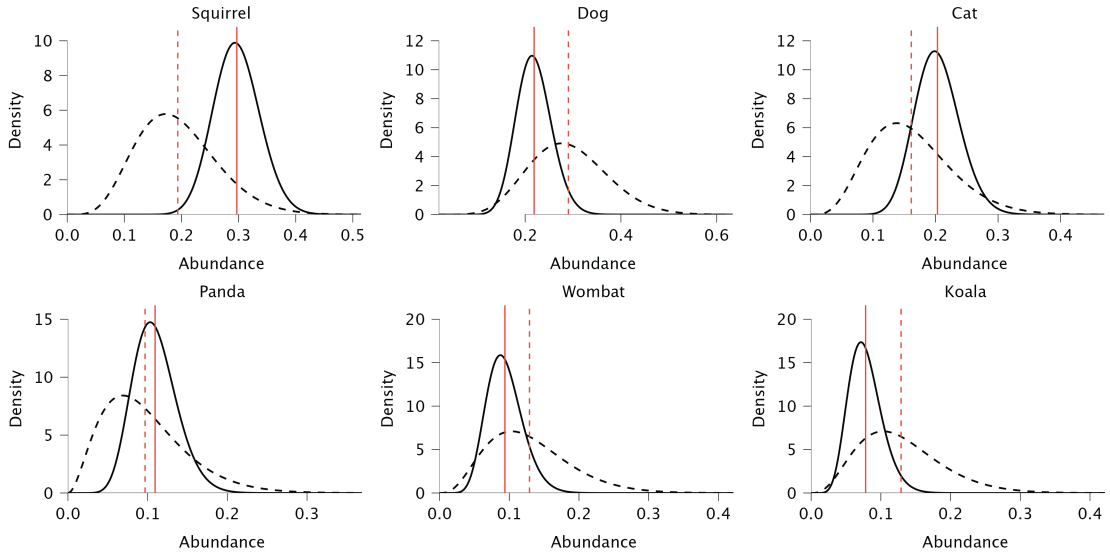


Figure 11

The distributions of the abundance of each species are described by Beta distributions, with prior (dotted lines), posterior (solid lines), and their means (red lines). The curves get more peaked as we collect more samples, reflecting that we become more certain with our estimates.

Table 12

The mean abundance of each observed species, along with the combined abundance of unknown species. Species with larger estimated abundance appear first in the columns

Hypothetical S	Squirrel	Dog	Cat	Panda	Wombat	Koala	Unknown
S = 6	0.297	0.219	0.203	0.109	0.094	0.078	0.000
S = 7	0.295	0.217	0.202	0.109	0.093	0.078	0.008
S = 8	0.292	0.215	0.200	0.108	0.092	0.077	0.015
S = 9	0.290	0.214	0.198	0.107	0.092	0.076	0.023
S = 10	0.288	0.212	0.197	0.106	0.091	0.076	0.030

practising calculating the means of abundance distributions.

Likelihood

The "Likelihood" section deals with the question of "how likely our data are given a hypothetical species count". To calculate the likelihood, we need the species abundance of each species. As the model assumes that there is an infinite number of animals on the island and each individual has the same probability of being included in the sample, the probability of capturing an individual of a given species is the mean of that species' abundance distribution. Calculating the likelihood of a hypothetical species count on the island is difficult, as we have to take into account whether we have encountered a species before. The mathematical details have been explained in the previous section and, again, this part is not mandatory for introductory students. Table 13 shows the logarithmic likelihood of our observed data for each hypothetical species count. As the likelihood is the product of a series of numbers smaller than 1, it shrinks rapidly with large samples. Thus, we strongly recommend that users transform the likelihood to a logarithmic scale in the user interface to avoid dealing with extremely small numbers.

Table 13

We recommend using the log likelihood, especially with large samples, because the likelihood shrinks quickly as samples accumulate.

Hypothetical S	Sample likelihood	Overall likelihood
6	−165.0	−205.4
7	−166.4	−206.6
8	−167.8	−208.1
9	−169.2	−209.8
10	−170.5	−211.5

Another important feature in this section is the pizza plots and Bayes factors. Users can enable this feature by ticking the "Bayes factor: compare likelihood" checkbox and moving some

hypothetical numbers of species in pairs from the left variable list to the right. Recall that the Bayes factor (*BF*) evaluates how much better, or worse, one hypothesis predicts the observed data than another, and students can easily interpret it as "the observed data are *BF* times more likely under one hypothesis than the other". As in the static module, we use pizza plots (see Figure 12 as an example) to visualise the Bayes factor. The interpretation of the pizza plots has been introduced before, and it remains unchanged here.

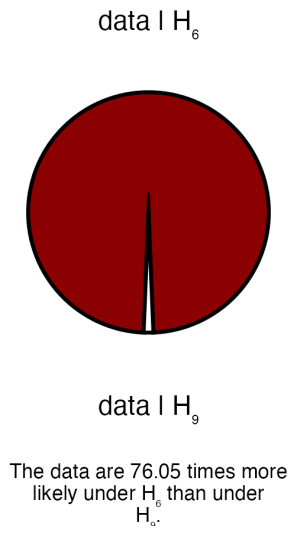


Figure 12

Our example data are much more surprising under the hypothesis that there are 9 different species on the fictional island than under the hypothesis that there are 6.

In classroom settings, we do not expect students to calculate the likelihood on their own, especially with large samples. Instead, our goal is to enable students to interpret the JASP output and make sense of the results qualitatively. Some example questions to think about are: (a) if we have already observed *s* species in our sample, which hypothetical species count is no longer possible; (b) if our sample size is large enough, which hypothesis is most supported by the observed data, and why; and (c) if we have observed only 5 animals belonging to different species, which hypothesis predicts the data best? These questions can enhance students' understanding of likelihood and can also stimulate discussions because the answers are sometimes counterintuitive.

Posterior Belief

The "posterior belief" section contains two parts: the posterior belief for how many different species there are on the fictional island (i.e., species count), and the posterior belief for the abundance of each of our observed species (i.e., species abundance).

For the species count posterior, our module follows the same structure as the static module. The first feature of this part is the belief update tables and plots. An example belief update table is shown in Table 14. In the table, we first show the prior for each of the hypothetical species counts specified previously and the relative likelihood, which is the likelihood of each hypothesis (calculated in the last section) divided by the likelihood of the hypothesis with the highest likelihood. Then, we showed the product of the prior and the likelihood, which is finally normalised to calculate the posterior probability for each hypothetical species count. An example belief update plot is shown in Figure 13. The bar plot is useful for comparing the posterior with the prior, and students can gain a better understanding of Bayes' rule by seeing how the observed data changed their belief.

Table 14

According to Bayes' rule, the posterior probability is the normalised product of the prior probability and the likelihood.

Species Count	Prior	Likelihood	Prior × Likelihood	Posterior
6	0.168	1.000	0.168	0.750
7	0.144	0.328	0.047	0.211
8	0.108	0.071	0.008	0.034
9	0.072	0.013	9.479×10^{-4}	0.004
10	0.043	0.002	9.768×10^{-5}	4.356×10^{-4}

Note. Likelihood values are relative to the hypothesis with the highest likelihood (H_6).

The second feature for the species count posterior is the evidence accumulation plot.

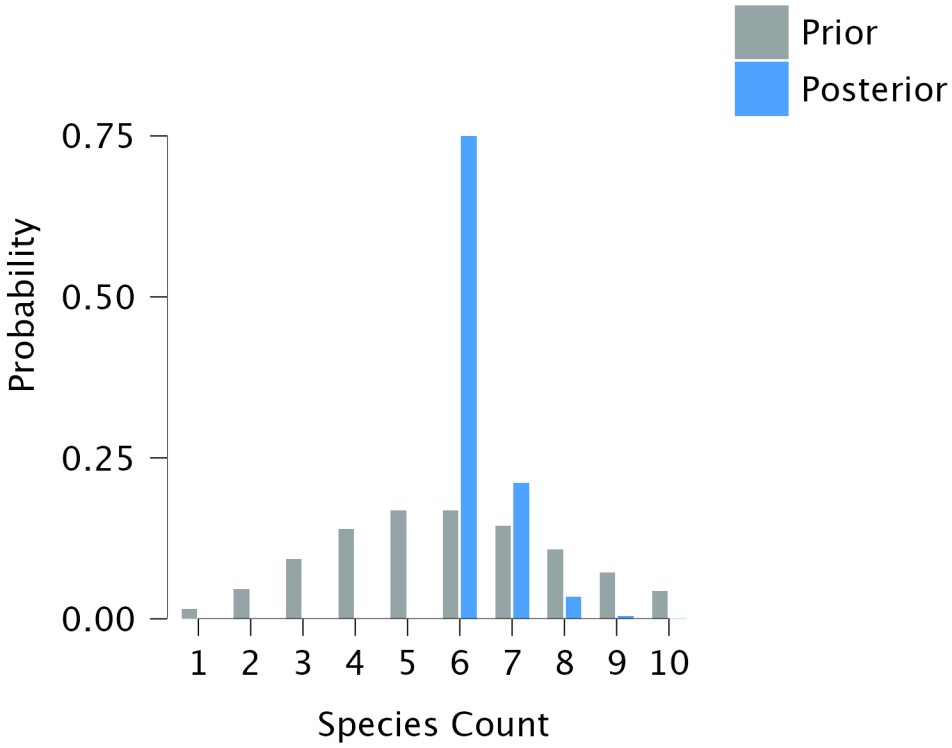


Figure 13

Having observed 6 different species in our sample, hypotheses with a species count lower than 6 become impossible, and the hypothesis that there are exactly 6 species on the fictional island has the highest posterior probability.

Again, as in the static module, users can enable this feature by ticking the "Plot" checkbox under "Evidence accumulation", and moving some hypotheses from the left to the right. As shown in the sample plot (Figure 14), each hypothesis specified is represented by a line with different colours on the plot, and the dots are the posterior probabilities after each sample. Unlike in the static module, the *x*-axis is now the number of individuals, instead of samples, so larger samples move the dots further to the right. The evidence accumulation plot allows a direct comparison of different hypotheses as we collect more data, and Bayes factors are obtainable by comparing the heights of the dots.

The third feature for the species count posterior is the posterior comparison table. This is also a feature of our static module. The main idea of this part is again to teach the odds form of

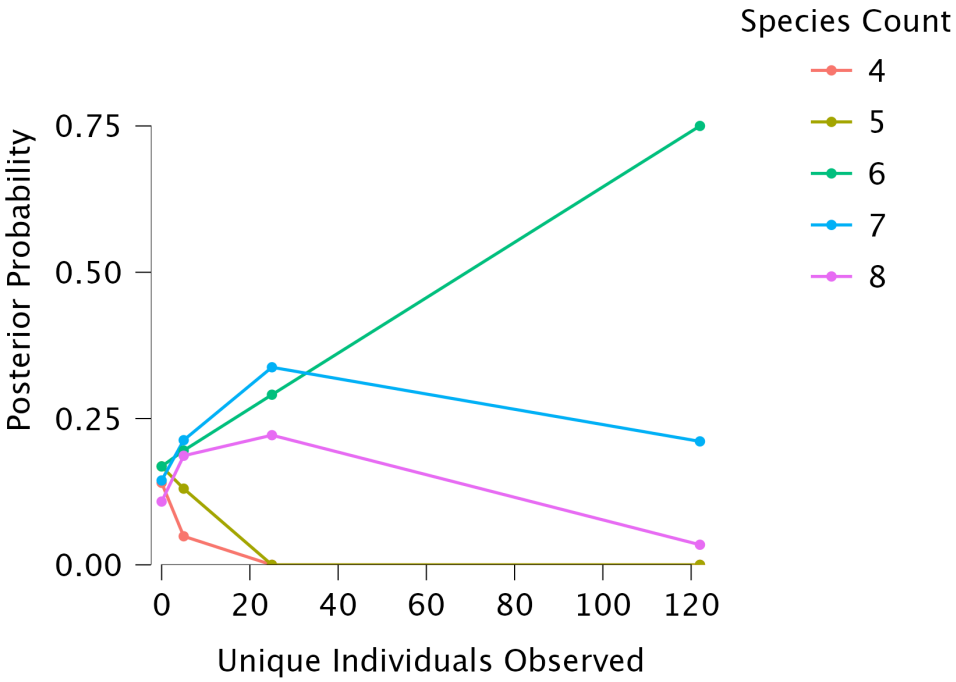


Figure 14

An evidence accumulation plot shows how our belief changes with the number of unique individuals in our samples. Note that dots represent our posterior belief after each sample and larger samples account for bigger jumps from the left to the right.

Bayes' rule. A posterior comparison table is available after checking the "Compare hypotheses" checkbox and specifying pairs of comparisons. In our example (Table 15), we specify two comparisons, which are H_6 vs. H_9 and H_7 vs. H_{10}. We can see that for both comparisons, the data are strongly in favour of the hypothesis with a smaller hypothetical species count. The odds form of Bayes' rule is convenient because the posterior odds are simply the product of the prior odds and the likelihood ratio, which directly reflects the essence of Bayes' rule — posterior is proportional to prior times likelihood.

For the species abundance posterior, we answer the question "what is the marginal distribution of the abundance of each species?". This differs from the previous "Abundance" section because in the previous section, the abundance of each species is conditioned on the hypothetical species count on the island. Here, we want to combine the distributions of each

Table 15

The posterior odds is the product of the prior odds and the likelihood ratio. In our example, the data overwhelmingly favour the hypotheses with smaller species counts (H_6 and H_7) in comparison to those with larger species counts (H_9 and H_{10}).

Hypothesis 1	Hypothesis 2	Prior odds	Likelihood ratio (BF)	Posterior odds
H_6	H_9	2.333	76.05	177.5
H_7	H_{10}	3.333	145.3	484.3

species' abundance under different hypotheses to make a single marginal distribution. The detailed calculations have been explained in the previous section, which essentially is to calculate the weighted average of different distributions, where the weights are the posterior probability of different hypotheses. In JASP, the species abundance table and plot are available. For the species abundance table (Table 16), users can choose between the mean, median, or mode to display, and it summarises the marginal distributions of each species by calculating the chosen statistics. For the species abundance plot (Figure 15), the software draws all the abundance distributions on a single graph. We recommend using the plot because it shows the full shape of the distributions, instead of just estimates.

In classroom settings, we expect students to qualitatively understand how to calculate the marginal distribution of each species' abundance from its abundance under each hypothetical species count, since it involves the basic idea of Bayesian model averaging. However, it is not mandatory for students to perform the calculations by hand. It is also helpful to compare the marginal abundance distribution with how often each species appears in the sample: the more individuals of a species we observe, the further right its marginal abundance distribution shifts.

Posterior Prediction

The final section of this module is "Posterior Prediction". Again, it predicts what we would observe if we were to see one more animal, given our posterior belief. By enabling the

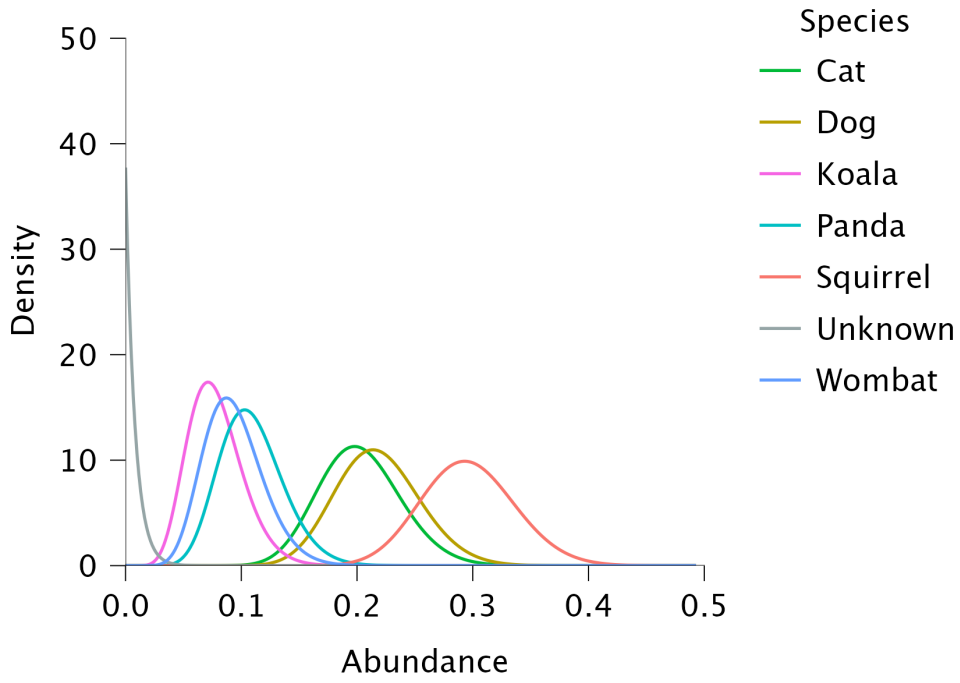


Figure 15

We calculated the marginal distribution of each species using Bayesian model averaging, where the marginal distribution is the weighted average of the distributions under each hypothesis.

Table 16

This table shows the means of the abundance distributions for each species.

Species	Marginal abundance
Squirrel	0.296
Dog	0.218
Cat	0.203
Panda	0.109
Wombat	0.094
Koala	0.078
Unknown	0.002

table and/or the plot checkbox, users can see the posterior predictive probability of each species summarised in a table or a plot. Both the table and the plot support customised settings, including displaying results by seen vs. unseen species or by individual species. An example posterior predictive table of seen vs. unseen species is shown in Table 17. Note that when the posterior prediction table is set to display individual species, users may find that the displayed values are exactly the same as those shown in Table 16. This follows directly from the model's assumptions: each individual has the same probability of entering our sample, so the probability of observing a particular species equals how well it is represented on the island (i.e., its abundance).

Table 17

In our example, the chances of observing an animal we have never seen before in the next observation are only about 0.002.

Category	Predicted probability
Seen species	0.998
Unseen species	0.002

Up to this point, we have finished our walk-through of the "Biodiversity Estimation (Full)" module. The following short discussion reflects on the design and mathematical model of the full module, and how it builds on the static module.

Discussion

The aim of the full module is to deepen students' understanding of the basic concepts of Bayesian statistics by making the model slightly more complicated and realistic. To accomplish this goal, we relaxed some assumptions made in the static module. Firstly, we no longer assume equal abundance across species where there are multiple species in a hypothesis. Instead, we estimate the abundance of species through our observed data. This approach may help us calculate the likelihood and thus the posterior more accurately and is therefore more realistic. Secondly, we no longer restrict the scenario to a fixed set of candidate species and have increased

the maximum number of species to 10.

Although we understand that relaxing those assumptions makes the material more difficult, we believe that these improvements are beneficial to students. In terms of biodiversity estimation, it is more common to estimate the abundance of species from a sample than to set a fixed and equal abundance for each species (Chao & Chiu, 2016). After all, there are always some rare species along with some dominant species on an island. In terms of statistics, estimating the marginal abundance of each species is essentially a Bayesian model averaging process, which is a significant approach in Bayesian statistics (Hinne et al., 2020; Hoeting et al., 1999). In addition, the full module actively introduces some common distributions to students. Since working with distributions instead of fixed values is common practice in research, students can benefit from practising with these common distributions in the module, becoming familiar with tools that are extremely useful in practice.

Despite having relaxed some assumptions from the static module, the full module still has some critical assumptions, making it different from real-life practices. Firstly, the full module still assumes that there is an infinite number of individual animals on the fictional island, which might slightly bias our posterior belief. As we have recorded the individuals we captured in each of the samples and can distinguish new individuals from recaptures, a more reasonable approach would be to estimate the total number of individuals based on the collected samples (Chao, 1984) and calculate the likelihood accordingly, which would be much more complicated. Secondly, when we estimate the species abundance, we use a uniform Dirichlet prior, which is reasonable in the scenario where no prior information is available. However, we could also allow for a customised prior where there are some dominant species with higher prior abundances. Thirdly, although we have increased the maximum possible total number of species to 10 in the full module, it is still a small number compared to what is happening in real-life biodiversity estimation because technically the number of different species in reality could be much greater than 10. However, we decided to keep these restrictions to avoid making the mathematical model so complicated that it overwhelms beginners.

The full module also contains user-friendly features that make it easier to navigate for both lecturers and students. Firstly, the full module closely follows the structure of the static module, where one section deals with a central concept in Bayesian statistics (i.e., prior, likelihood, posterior, etc.). The only difference is that we introduce the species abundance before calculating the likelihood, because it is available once we have our observed data, and we need the abundance to calculate the likelihood. Secondly, just like the static module, the tables are designed to be as intuitive as possible, where the calculations are divided into steps that move across columns. Thus, walking through the tables achieves the goal of introducing the seemingly complicated formulas. Thirdly, the plots are carefully designed to provide visualisations with the fewest elements possible so users can grasp the central idea easily.

In addition to the considerations that went into designing the module, we believe that the effort of the lecturers matters as well. In the example walk-through section, we have introduced some example course activities, which we think might be helpful. Lecturers can also come up with some other ideas for courses or exercises. The principle of our module and course design is to focus more on understanding the general ideas of Bayesian statistics and less on mathematical calculations. Therefore, it is more important to spend time helping students understand the JASP outputs than to give students raw data and ask them to perform extensive calculations. Also, as we have mentioned in the previous sections, some of the mathematical details may be too advanced for beginners, and focusing too much on those might overwhelm them.

General Discussion

The goal of our project is to develop example Bayesian teaching modules in JASP for beginners. We designed two modules under the same framework: estimating the biodiversity of a fictional island, namely how many different species there are on the island and what the abundance of each species is. In the following parts, we will first evaluate our project in terms of our module design and Bayesian statistics itself, and then conclude with some limitations and future directions that might inform future work.

Reflections on the Two-Module Design

As we have mentioned in the introduction, a reason why young students receive limited Bayesian education is that there are not enough ready-to-use teaching and learning materials. Our modules serve as a valid attempt to bridge this gap, as they fulfil the criteria that we put forward in the introduction.

Firstly, our modules feature an example that is both appealing to young students and practically relevant. As suggested by Lovett and Greenhouse (2000), an effective approach to teaching statistics is to provide examples that may stimulate students' interest in the subject. Both of our modules have a common background scenario, which is to estimate the biodiversity on a fictional island. This topic is easy to understand and is considered appealing to young students (Bermudez & Lindemann-Matthies, 2020), who are familiar with similar topics and have the curiosity to explore nature. Furthermore, the example also benefits students because investigating biodiversity is a common approach in life sciences, and students may have done similar investigations themselves or may need to do so in the future. Thus, we chose biodiversity estimation as our example, since it benefits students in terms of both statistics and other aspects of their studies.

Secondly, our modules have clear demonstrations and visualisations, which are enabled by JASP, an open-access, easy-to-use software. Many researchers have highlighted the use of software in statistical education to help students make sense of abstract knowledge (Bromage et al., 2022; Garfield, 1995). Our modules make the most of JASP's advantages. As seen in the previous sections, we have designed and programmed various tables and plots to make them easier for students to understand. For example, by using pizza plots, Bayes factors are no longer just numbers; they become something that users can actually imagine and feel. Similarly, by using belief update tables, Bayes' rule is no longer a seemingly complicated formula, but something unpacked through a series of comprehensible steps. Users do not need any knowledge of coding to interact with the JASP interface, and explanatory texts are available for each section, which navigate the users through the software and Bayesian statistics. Furthermore, throughout

this report, we have introduced ways to interpret these tables and plots (e.g., the pizza-dipping analogy for Bayes factors), which lecturers can draw on directly when thinking about how to explain the knowledge in class. In summary, the demonstrations and visualisations in our modules make it easy for both students to learn and lecturers to teach.

Thirdly, our project contains two modules that are closely linked to each other and suitable for students of different levels, offering a connected learning path for students to progress. As discussed above, Bayesian courses are limited to advanced students due to a lack of introductory-level materials (Dogucu & Hu, 2022; Hoegh, 2020). The static module in our project is built for absolute beginners to Bayesian statistics, even without strong mathematical foundations. The static module involves many assumptions that simplify the calculations, and knowledge is explained one concept at a time, to minimise cognitive load. As specified in our learning objectives, the goal of our static module is not to teach complicated calculations, but to help students become familiar with the general idea of Bayesian statistics. Then we have the full module, which makes the calculations slightly more complicated by adding the estimation of species abundance. As students use the full module, they will also gain hands-on experience working with common distributions in statistics (e.g., uniform, Poisson, and Beta distributions), which may benefit their future studies. We believe that the connection between these two modules offers a steady learning path for absolute beginners, which is consistent with the principle of scaffolding, in which students build competence and confidence through progressing from simpler to more complex tasks (Wood et al., 1976).

In summary, our JASP modules successfully fulfilled the criteria we put forward in the introduction: they feature an engaging example, provide clear demonstrations and visualisations, and are suitable for students ranging from absolute beginners to those with some experience. Through engaging with the two modules, users may not learn everything about biodiversity estimation or complicated probability calculations, but they may gradually build up their knowledge of how Bayesian statistics works. Furthermore, thoughtful students can also get some general ideas on how to think like a Bayesian and reflect on the models independently, which is

what we really care about.

Revisiting the Intuitiveness of Bayesian Statistics

As discussed in the introduction, the current study, along with some previous research (Etz & Vandekerckhove, 2018; Sedgwick et al., 2022; Wagenmakers & Matzke, 2024; Wagenmakers et al., 2018), argues that Bayesian statistics is more intuitive than traditional frequentist statistics. Having developed these teaching modules, we would like to revisit how the intuitiveness of Bayesian statistics is manifested in our modules and what it means to students.

Firstly, our modules, along with Bayesian statistics itself, reflect the intuition that the posterior is a combination of prior belief and likelihood, which echoes the everyday experience of learning from the past to inform the future. One of the main differences between Bayesian and frequentist statistics is that Bayesian statistics require a prior belief to be specified before observing the data (Baig, 2020; Wagenmakers et al., 2018). The prior serves as the starting point for our belief update, and different prior beliefs will bring us to various posterior beliefs. In our modules, we encourage users to try different priors, including extreme ones, to see how much the resulting posterior beliefs can differ. This aligns perfectly with students' everyday experience that two people who have seen the same thing do not necessarily hold the same belief afterward. Then, there is likelihood, which tells us how likely our observed data are under different hypotheses. In our static module, the "likelihood" section follows closely after the "data" section and is independent of prior belief. In other words, this is the part that is objective. After preparing these two ingredients, the posterior is obtained directly by combining the prior and the likelihood through simple calculations. This design not only reduces the cognitive load of students (as discussed in the previous part), but it also aligns with the natural process of belief update in everyday life.

Secondly, our modules showcase the intuitiveness of Bayesian statistics by making it possible to directly compare the likelihood and the posterior probability of rival hypotheses, rather than testing a single hypothesis on its own. In frequentist statistics, we make an inference whether the alternative hypothesis is true based on how unlikely the data are assuming that the

null hypothesis is true (Greenland et al., 2016). This logic is complicated for beginners, and the null hypothesis is not given a fair chance to gain support. By contrast, Bayesian statistics allow direct comparisons of rival hypotheses, where we specify the prior, calculate the likelihood, and the posterior for each of them (Dienes & Mclatchie, 2018; Etz & Vandekerckhove, 2018; Wagenmakers et al., 2018). In our modules, we have also highlighted this characteristic of Bayesian statistics in several places. For example, in the likelihood section, we introduced pizza plots to demonstrate Bayes factors, and these pizza plots are also used in the posterior section in the static module where the odds form of Bayes' rule comes into play. Note that although previous studies and textbooks have used pizza plots to show Bayes factors (Wagenmakers & Matzke, 2024), to our knowledge, our module is among the first to use them to show the prior odds and the posterior odds. We would like to express the message that everything is directly comparable in Bayesian statistics, not just the likelihood. The odds form of Bayes' rule is a powerful tool, and becoming familiar with it can benefit students significantly. Additionally, we have shown the ability of Bayesian statistics to compare hypotheses directly through evidence accumulation plots, where each hypothesis is represented by a separate line, and users can see how evidence changes as data accumulate.

Thirdly, our modules demonstrate that results of Bayesian statistics can be written in plain and simple language. Researchers have found that some of the concepts in traditional frequentist statistics might be difficult for students and even researchers to understand (Greenland et al., 2016), causing people to misinterpret them. By contrast, Bayesian statistics may be a good alternative because its results are simpler to interpret (Wagenmakers & Matzke, 2024). This difference is also shown in our modules, as we included simple and clear explanations of the JASP outputs in various places. For example, all Bayes factors can be interpreted in one simple and intuitive sentence: "The data are *BF* times more likely under one hypothesis than the other.", and we can explain the prior odds and posterior odds in similar ways. We argue that simple explanations are difficult to achieve for frequentist statistics, as its logic is much more complicated. Lecturers may also find that students become familiar with Bayesian statistics more

quickly and make fewer mistakes due to its simplicity.

However, rather than being too optimistic, we would like to point out that certain boundaries apply regarding the teaching of Bayesian statistics. Throughout our modules, we have focused on the theoretical idea of Bayesian statistics and have been using the analogy of "our present belief is the product of our past belief and the evidence from data" to help students make sense of Bayes' rule. However, curious students may wonder why this law holds, which involves the mathematical derivation of Bayes' rule by using probability theory. Although we have introduced the derivation in this report, the process is not mandatory in our modules in order to avoid overwhelming introductory students. The same problem occurs when calculating the likelihood, which is in fact a complicated mathematical process. The message here is that although Bayesian statistics is intuitive to make sense of, there are deep mathematical foundations behind the scenes, which we tried not to focus on in our modules. While introductory students will benefit from our modules by grasping the basic idea of Bayesian statistics, we still recommend that lecturers tailor the course design to the designated audience so that students can get the most out of our modules.

Limitations and Future Directions

Although our project carries theoretical and practical significance, it also has some notable limitations, which may also inform future research.

Firstly, in terms of the mathematical models we used in our modules, they are not realistic enough for real-life biodiversity estimation. As we have discussed before, we specified multiple assumptions in the static module to simplify mathematical calculations. In the full module, we relaxed some of those assumptions to make the model more realistic. However, the full module is still simpler compared to real-life practices. For example, the full module assumes that the maximum possible number of species on an island is 10, which is not likely because the number could greatly exceed 10; the full module assumes that there are infinite individuals on the island, which may also slightly bias our estimation because the number of animals on the island is always finite; the full module also did not allow users to specify priors for species abundance, but used a

uniform prior instead, which is a default setting when there is no prior information, and may not be enough for users with expectations of species abundance a priori. We would like to emphasise that these assumptions and limitations were deliberate choices made to avoid overwhelming beginners with complicated mathematical calculations. However, when engaging with our modules, lecturers should keep in mind these limitations and view our modules more as teaching material for Bayesian statistics, rather than for serious biodiversity estimation. For advanced students, these modules would be a good starting point for more complicated scenarios, and they can also benefit from reflecting on the limitations of the mathematical models behind our modules.

Future work can further relax some of the assumptions from our full module, and build some more complicated modules on top of the two modules in our project. Among the above assumptions, the first, regarding the limit on the maximum number of species, is the easiest to tackle by using convergent series (Wagenmakers & Matzke, 2024) to specify a prior with an infinite hypothetical species count on the island. For the second assumption mentioned above, a better approach would be to estimate the total number of individuals on the island through the ratio of new and recaptured individuals in our samples, which is a common practice in biology (Chao, 1984; Chao & Chiu, 2016). For the last assumption, a main challenge when implementing priors for species abundance would be to avoid making the user interface too crowded and also to convey the idea of Bayesian model averaging to students, especially those at an introductory level.

Secondly, in terms of the effectiveness of our modules, it has not been examined through real-life classroom teaching. Our goal for this project is to develop ready-to-use teaching materials for Bayesian statistics that are suitable for beginners. At this point, we have fulfilled this goal, and our modules are designed with both introductory students and lecturers in mind. For students, we have implemented explanatory texts for each section of our static module and have made various assumptions to simplify mathematical calculations in both modules, so as to avoid overwhelming beginners and help them focus on general ideas of Bayesian statistics. For lecturers, we have designed clear and informative graphs throughout our modules and have

mentioned some sample class designs in this report. However, since these are teaching modules, we believe that the best way to test whether they are effective enough is to put them into actual classroom teaching.

Future studies can substantially contribute to this project and the field by doing some field experiments using these modules to teach Bayesian statistics to students of different levels, especially those who are absolute beginners. Through documenting detailed class design and real-life feedback from students, we can see which part of the framework beginners struggle with the most, and improve our modules accordingly. Furthermore, future studies can also compare the learning outcomes of our modules with courses featuring traditional frequentist statistics. We consider this effort extremely valuable because, on the one hand, it provides evidence as to whether Bayesian statistics is more intuitive for beginners than its counterpart, and, on the other hand, it helps promote Bayesian statistics and increases Bayesian literacy among young students, which may be helpful for their future.

Conclusion

Our project developed two JASP example modules to teach Bayesian statistics to beginners. Both modules feature the same example of estimating the number of different species on a fictional island. The Biodiversity Estimation (Static) module aims to introduce the basic Bayesian concepts as clearly as possible, and the Biodiversity Estimation (Full) module builds upon the static module by estimating the abundance of each species from our samples, making it more realistic. Theoretically, our project demonstrates the intuitiveness of Bayesian statistics and contributes to the debate between Bayesian and frequentist statistics. Practically, our modules, along with the report, fill the gap of limited teaching and learning materials for Bayesian statistics that are suitable for absolute beginners, and may assist lecturers in designing courses. Taken together, we consider our project a meaningful step in promoting Bayesian statistics, increasing Bayesian literacy, and empowering students for their future scientific research.

Module Availability

Our project contains two modules, Biodiversity Estimation (Static) and Biodiversity Estimation (Full). We will make both modules available in the next version of JASP, which is free, open-access statistical software (JASP Team, 2026). Both modules will be placed under "Learn Bayes", which users may have to install by clicking the "plus" button on the top-right corner of the software. We welcome all kinds of feature requests and bug reports through the JASP forum.

Declaration of AI

The project used generative AI (Claude 5.2) only to assist with bug-fixes in the process of JASP programming, and to correct grammar mistakes in the revision phase. No AI was used in any other circumstances throughout the project.

Acknowledgement

The project would never have been possible without teamwork. As the intern, I would like to thank my supervisor, Prof. Dr. Eric-Jan Wagenmakers, for bringing up this exciting and meaningful project and offering valuable suggestions as we developed the JASP modules. I would also like to thank my daily supervisor, MSc. Henrik Godmann, for his support, for thinking through problems together with me, and for stimulating thought-provoking discussions throughout the development of our project.

References

Altman, D. G., & Bland, J. M. (1995). Absence of evidence is not evidence of absence. *BMJ*, *311*(7003), 485. https://doi.org/10.1136/bmj.311.7003.485

Baig, S. A. (2020). Bayesian Inference: An Introduction to Hypothesis Testing Using Bayes Factors. *Nicotine & Tobacco Research*, *22*(7), 1244–1246. https://doi.org/10.1093/ntr/ntz207

Bermudez, G., & Lindemann-Matthies, P. (2020). "What Matters Is Species Richness"—High School Students' Understanding of the Components of Biodiversity. *Research in Science Education*, *50*, 2159–2187. https://doi.org/10.1007/s11165-018-9767-y

Berry, D. A. (1997). Teaching Elementary Bayesian Statistics with Real Applications in Science. *The American Statistician*, *51*(3), 241–246. https://doi.org/10.2307/2684895

Bon, J. J., Bretherton, A., Buchhorn, K., Cramb, S., Drovandi, C., Hassan, C., Jenner, A. L., Mayfield, H. J., McGree, J. M., Mengersen, K., Price, A., Salomone, R., Santos-Fernandez, E., Vercelloni, J., & Wang, X. (2023). Being Bayesian in the 2020s: Opportunities and challenges in the practice of modern applied Bayesian statistics. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *381*(2247), 20220156. https://doi.org/10.1098/rsta.2022.0156

Bromage, A., Pierce, S., Reader, T., & Compton, L. (2022). Teaching statistics to non-specialists: Challenges and strategies for success. *Journal of Further and Higher Education*, *46*(1), 46–61. https://doi.org/10.1080/0309877X.2021.1879744

Burnham, K. P., & Overton, W. S. (1978). Estimation of the Size of a Closed Population when Capture Probabilities vary Among Animals. *Biometrika*, *65*(3), 625–633. https://doi.org/10.2307/2335915

Chao, A. (1984). Nonparametric Estimation of the Number of Classes in a Population. *Scandinavian Journal of Statistics*, *11*(4), 265–270. https://www.jstor.org/stable/4615964

Chao, A., & Chiu, C.-H. (2016). Species Richness: Estimation and Comparison. In *Wiley StatsRef: Statistics Reference Online* (pp. 1–26). John Wiley & Sons, Ltd. https://doi.org/10.1002/9781118445112.stat03432.pub2

Depaoli, S., Rus, H. M., Clifton, J. P., van de Schoot, R., & Tiemensma, J. (2017). An introduction to Bayesian statistics in health psychology. *Health Psychology Review*, *11*(3), 248–264. https://doi.org/10.1080/17437199.2017.1343676

Dienes, Z., & Mclatchie, N. (2018). Four reasons to prefer Bayesian analyses over significance testing. *Psychonomic Bulletin & Review*, *25*(1), 207–218. https://doi.org/10.3758/s13423-017-1266-z

Dogucu, M., & Hu, J. (2022). The Current State of Undergraduate Bayesian Education and Recommendations for the Future. *The American Statistician*, *76*(4), 405–413. https://doi.org/10.1080/00031305.2022.2089232

Dunn, K. (2014). Why Wait? The Influence of Academic Self-Regulation, Intrinsic Motivation, and Statistics Anxiety on Procrastination in Online Statistics. *Innovative Higher Education*, *39*(1), 33–44. https://doi.org/10.1007/s10755-013-9256-1

Etz, A., & Vandekerckhove, J. (2018). Introduction to Bayesian Inference for Psychology. *Psychonomic Bulletin & Review*, *25*(1), 5–34. https://doi.org/10.3758/s13423-017-1262-3

Ferrari, D. (2022). Teaching Bayesian Statistics. *PS: Political Science & Politics*, *55*(1), 230–235. https://doi.org/10.1017/S1049096521001244

Garfield, J. (1995). How Students Learn Statistics. *International Statistical Review / Revue Internationale de Statistique*, *63*(1), 25–34. https://doi.org/10.2307/1403775

Gelman, A., Carlin, J. B., Stern, H. S., & Rubin, D. B. (2003, July). *Bayesian Data Analysis* (2nd ed.). Chapman; Hall/CRC. https://doi.org/10.1201/9780429258480

Greenland, S., Senn, S. J., Rothman, K. J., Carlin, J. B., Poole, C., Goodman, S. N., & Altman, D. G. (2016). Statistical tests, P values, confidence intervals, and power: A guide to misinterpretations. *European Journal of Epidemiology*, *31*, 337–350. https://doi.org/10.1007/s10654-016-0149-3

Hanna, D., & Dempster, M. (2009). The Effect of Statistics Anxiety on Students' Predicted and Actual Test Scores. *The Irish Journal of Psychology*, *30*(3-4), 201–209. https://doi.org/10.1080/03033910.2009.10446310

Henriques, A. (2016). Students' Difficulties in Understanding of Confidence Intervals. In D. Ben-Zvi & K. Makar (Eds.), *The Teaching and Learning of Statistics: International Perspectives* (pp. 129–138). Springer International Publishing. https://doi.org/10.1007/978-3-319-23470-0_18

Hinne, M., Gronau, Q. F., van den Bergh, D., & Wagenmakers, E.-J. (2020). A Conceptual Introduction to Bayesian Model Averaging. *Advances in Methods and Practices in Psychological Science*, *3*(2), 200–215. https://doi.org/10.1177/2515245919898657

Hoegh, A. (2020). Why Bayesian Ideas Should Be Introduced in the Statistics Curricula and How to Do So. *Journal of Statistics Education*, *28*(3), 222–228. https://doi.org/10.1080/10691898.2020.1841591

Hoeting, J. A., Madigan, D., Raftery, A. E., & Volinsky, C. T. (1999). Bayesian model averaging: A tutorial. *Statistical Science*, *14*(4), 382–417. https://doi.org/10.1214/ss/1009212519

Hunt, B. W., Mari, T., Knibb, G., Christiansen, P., & Jones, A. (2023). Statistics anxiety and predictions of exam performance in UK psychology students. *PLOS ONE*, *18*(8), e0290467. https://doi.org/10.1371/journal.pone.0290467

JASP Team. (2026). JASP (Version 0.96.0)[Computer software]. https://jasp-stats.org/

Jeffreys, H. (1961). *The Theory of Probability* (Third Edition). Oxford University Press.

Johnson, V. E. (2013). Revised standards for statistical evidence. *Proceedings of the National Academy of Sciences*, *110*(48), 19313–19317. https://doi.org/10.1073/pnas.1313476110

Kolmogorov, A. N. (1933). *Grundbegriffe der wahrscheinlichkeitsrechnung*. Springer.

Kruschke, J. (2014, January). *Doing Bayesian data analysis: A tutorial with R, JAGS, and Stan, second edition* [Pages: 759]. Academic Press. https://doi.org/10.1016/B978-0-12-405888-0.09999-2

Lethbridge, L. E., Marshall, A. D., & Jauch, M. (2024). Attitudes towards statistics and statistical performance: A mediation model of statistics anxiety and academic procrastination. *British Journal of Educational Psychology*, *94*(4), 1208–1223. https://doi.org/10.1111/bjep.12714

Lovett, M. C., & Greenhouse, J. B. (2000). Applying Cognitive Theory to Statistics Instruction. *The American Statistician*, *54*(3), 196–206. https://doi.org/10.2307/2685590

MacArthur, K. R., & Santo, J. B. (2023). A Multi-Level Analysis of the Effects of Statistics Anxiety/Attitudes on Trajectories of Exam Scores. *Journal of Statistics and Data Science Education*, *31*(1), 102–112. https://doi.org/10.1080/26939169.2022.2093805

McElreath, R. (2020, March). *Statistical Rethinking: A Bayesian Course with Examples in R and STAN* (2nd ed.). Chapman; Hall/CRC. https://doi.org/10.1201/9780429029608

Meng, X.-L. (2009). Desired and Feared—What Do We Do Now and Over the Next 50 Years? *The American Statistician*, *63*(3), 202–210. https://www.jstor.org/stable/25652268

Nuzzo, R. (2014). Scientific method: Statistical errors. *Nature*, *506*(7487), 150–152. https://doi.org/10.1038/506150a

Onwuegbuzie, A. J., & Wilson, V. A. (2003). Statistics Anxiety: Nature, etiology, antecedents, effects, and treatments–a comprehensive review of the literature. *Teaching in Higher Education*, *8*(2), 195. https://doi.org/10.1080/1356251032000052447

Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., Mensink, L., Nak, J., De Ruiter, J. P., Sarafoglou, A., Siepe, B. S., Arena, G., Akrong, E., Aust, F., van den Bergh, D., & Wagenmakers, E.-J. (2026). A methodological metamorphosis: The rapid rise of Bayesian inference and open science practices in psychology [Preprint]. https://doi.org/10.31234/osf.io/ck3js_v2

Puklek Levpušček, M., & Cukon, M. (2020). That Old Devil Called 'Statistics': Statistics Anxiety in University Students and Related Factors. *Center for Educational Policy Studies Journal*. https://doi.org/10.26529/cepsj.826

Rindskopf, D. (2020). Overview of Bayesian Statistics. *Evaluation Review*, *44*(4), 225–237. https://doi.org/10.1177/0193841X19895623

Schoot, R., Depaoli, S., King, R., Kramer, B., Märtens, K., Tadesse, M., Vannucci, M., Gelman, A., Veen, D., Willemsen, J., & Yau, C. (2021). Bayesian statistics and modelling. *Nature Reviews Methods Primers*, *1*. https://doi.org/10.1038/s43586-020-00001-2

Sedgwick, P. M., Hammer, A., Kesmodel, U. S., & Pedersen, L. H. (2022). Current controversies: Null hypothesis significance testing. *Acta Obstetricia et Gynecologica Scandinavica*, *101*(6), 624–627. https://doi.org/10.1111/aogs.14366

Silva, P. N., & Sarnecka, B. W. (2025). What Do Your Students Struggle with? A Survey of Statistics Instructors. *Journal of Statistics and Data Science Education*, *0*(0), 1–12. https://doi.org/10.1080/26939169.2025.2455560

Stern, H. S. (2016). A Test By Any Other Name: P-values, Bayes Factors and Statistical Inference. *Multivariate behavioral research*, *51*(1), 23–29. https://doi.org/10.1080/00273171.2015.1099032

Szucs, D., & Ioannidis, J. P. A. (2017). When Null Hypothesis Significance Testing Is Unsuitable for Research: A Reassessment. *Frontiers in Human Neuroscience*, *11*, 390. https://doi.org/10.3389/fnhum.2017.00390

Tishkovskaya, S., & Lancaster, G. A. (2012). Statistical Education in the 21st Century: A Review of Challenges, Teaching Innovations and Strategies for Reform. *Journal of Statistics Education*, *20*(2). https://doi.org/10.1080/10691898.2012.11889641

Trassi, A. P., Leonard, S. J., Rodrigues, L. D., Rodas, J. A., & Santos, F. H. (2022). Mediating factors of statistics anxiety in university students: A systematic review and meta-analysis. *Annals of the New York Academy of Sciences*, *1512*(1), 76–97. https://doi.org/10.1111/nyas.14746

van de Schoot, R., Winter, S. D., Ryan, O., Zondervan-Zwijnenburg, M., & Depaoli, S. (2017). A systematic review of Bayesian articles in psychology: The last 25 years. *Psychological Methods*, *22*(2), 217–239. https://doi.org/10.1037/met0000100

Wagenmakers, E.-J., Love, J., Marsman, M., Jamil, T., Ly, A., Verhagen, J., Selker, R., Gronau, Q. F., Dropmann, D., Boutin, B., Meerhoff, F., Knight, P., Raj, A., van Kesteren, E.-J., van Doorn, J., Šmíra, M., Epskamp, S., Etz, A., Matzke, D., … Morey, R. D. (2018). Bayesian inference for psychology. Part II: Example applications with JASP. *Psychonomic Bulletin & Review*, *25*(1), 58–76. https://doi.org/10.3758/s13423-017-1323-7

Wagenmakers, E.-J., & Matzke, D. (2024, August). *Bayesian inference from the ground up: The Theory of Commonsense*. Retrieved March 18, 2026, from https://surtil.com/2024/08/13/bayesian-inference-from-the-ground-up/

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA Statement on p-Values: Context, Process, and Purpose. *The American Statistician*, *70*(2), 129–133. https://doi.org/10.1080/00031305.2016.1154108

Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, *17*(2), 89–100. https://doi.org/10.1111/j.1469-7610.1976.tb00381.x